ENGINEERING CHEMISTRY LAB

Engineering Chemistry Lab is a course designed for engineering students to provide them with practical knowledge and hands-on experience in the field of chemistry. The course aims to equip students with the necessary skills to perform experiments and analyze data in a laboratory setting.

Some of the key objectives of the Engineering Chemistry Lab course are: - To provide students with an understanding of the basic principles and techniques used in chemistry experiments. - To enable students to perform experiments safely and accurately, and to analyze and interpret the data obtained. - To develop students' ability to work independently and in teams, and to communicate their findings effectively.

The course typically covers a range of topics, including: - Acid-base titrations - Redox titrations - Gravimetric analysis - Spectrophotometry - Chromatography - Thermochemistry - Electrochemistry

In the Engineering Chemistry Lab, students are expected to: - Follow all safety rules and regulations - Maintain a clean and organized workspace - Keep accurate and detailed records of their experiments - Work collaboratively with their peers - Communicate their findings clearly and concisely

Overall, the Engineering Chemistry Lab course provides students with a solid foundation in the practical aspects of chemistry, and prepares them for further study and research in the field.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance the ability to apply theoretical concepts to practical situations.
4. To encourage independent learning and research.
5. To foster effective communication and collaboration skills.
6. To prepare students for further academic or professional pursuits in the field.
7. To promote ethical and responsible behavior in the discipline.
8. To instill a lifelong love of learning and intellectual curiosity.

1. To enable the students to understand about the fundamental concepts of analytical instruments:

- Analytical instruments are tools used to measure, analyze, and quantify the physical and chemical properties of substances.
- These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering.
- Some common types of analytical instruments include spectrometers, chromatographs, and microscopes.

- Analytical instruments are essential for scientific research, quality control, and product development.
- Understanding the fundamental concepts of analytical instruments is important for students who wish to pursue careers in science or engineering.
- By learning about the principles and applications of analytical instruments, students can develop the skills and knowledge necessary to use these tools effectively in their future work.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water.

- **Chloride content analysis:** Chloride ions are one of the major inorganic anions in water and wastewater. The analysis of chloride content in water can be done by titration with silver nitrate using potassium chromate as an indicator. This method is known as the Mohr method. Another method for chloride analysis is the potentiometric titration using a silver ion-selective electrode.

- **Hardness analysis:** Hardness in water is caused by the presence of multivalent metallic cations, such as calcium and magnesium. The analysis of water hardness can be done by titration with a chelating agent, such as ethylenediaminetetraacetic acid (EDTA), using a suitable indicator.

- **Alkalinity analysis:** Alkalinity in water is a measure of its capacity to neutralize acids. It is primarily due to the presence of bicarbonate, carbonate, and hydroxide ions. The analysis of water alkalinity can be done by titration with a standard acid solution, such as sulfuric acid, using a suitable indicator.

These are some of the methods used to analyze the chloride content, hardness, and alkalinity of water. It is important for students to understand these methods as they provide valuable information about the quality of water.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration. A pH of 7 is considered neutral, while a pH less than 7 is acidic and a pH greater than 7 is basic.
- **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the liquid molecules. Surface tension can be measured using a number of methods, including the drop weight method and the maximum bubble pressure method.
- **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. Viscosity can be measured using a number of methods, including the capillary tube method and the rotating cylinder method.

These properties are important in many fields, including chemistry, physics, and engineering. Understanding how to measure and interpret these properties can help students better understand the behavior of liquids in various situations.

4. To enable the students to understand about the preparation of different resins.

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to consider when preparing different resins:

1. **Types of resins:** There are several types of resins, including natural resins, synthetic resins, and modified natural resins. Each type of resin has its own unique properties and preparation methods.
2. **Raw materials:** The raw materials used in the preparation of resins vary depending on the type of resin being prepared. For example, natural resins are obtained from plants, while synthetic resins are prepared from chemicals.
3. **Preparation methods:** The methods used to prepare resins vary depending on the type of resin being prepared. Some common methods include polymerization, condensation, and addition reactions.
4. **Safety precautions:** The preparation of resins often involves the use of chemicals and high temperatures. It is important to follow safety precautions, such as wearing protective equipment and working in a well-ventilated area.
5. **Quality control:** The quality of the prepared resin is important for its intended use. Quality control measures, such as testing the physical and chemical properties of the resin, should be implemented to ensure that the resin meets the desired specifications.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol by Conventional and Green Route

Adipic acid and paracetamol are two important organic compounds that can be synthesized through both conventional and green routes. Here, we will discuss the differences between these two methods of synthesis.

Adipic Acid

Adipic acid is an important dicarboxylic acid that is used in the production of nylon. It can be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of adipic acid involves the oxidation of cyclohexane or cyclohexanol with nitric acid. This process produces large amounts of nitrous oxide, a potent greenhouse gas.

Green Route

The green route for the synthesis of adipic acid involves the oxidation of cyclohexene with hydrogen peroxide in the presence of a tungsten catalyst. This process produces water as the only byproduct, making it a more environmentally friendly option.

Paracetamol

Paracetamol is a widely used analgesic and antipyretic drug. It can also be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of paracetamol involves the nitration of phenol to produce 4-nitrophenol, which is then reduced to 4-aminophenol. This intermediate is then acetylated to produce paracetamol.

Green Route

The green route for the synthesis of paracetamol involves the direct acetylation of 4-aminophenol with acetic anhydride in the presence of a solid acid catalyst. This process produces fewer byproducts and is considered to be more environmentally friendly.

In conclusion, the synthesis of organic compounds such as adipic acid and paracetamol can be achieved through both conventional and green routes. The green routes are generally considered to be more environmentally friendly due to the reduced production of harmful byproducts.

LIST OF EXPERIMENTS

1. **Milgram Experiment**: A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **Stanford Prison Experiment**: A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **Pavlov's Dogs**: A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
4. **Asch Conformity Experiment**: A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect**: A study on the effect of changes in the work environment on productivity, conducted by Elton Mayo in the 1920s.

6. **Bobo Doll Experiment**: A study on the effects of observed aggression, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment**: A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **The Marshmallow Test**: A study on delayed gratification, conducted by Walter Mischel in the 1960s.
9. **The Invisible Gorilla**: A study on inattentional blindness, conducted by Christopher Chabris and Daniel Simons in 1999.
10. **The Trolley Problem**: A thought experiment in ethics, first introduced by Philippa Foot in 1967.

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of an instrument or apparatus by comparing its measurements with a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

Here are some key points to consider when calibrating analytical equipment and apparatus:

- Calibration should be performed regularly to ensure the accuracy of the instrument.
- The frequency of calibration will depend on factors such as the type of instrument, its usage, and the environment in which it is used.
- Calibration should be performed by trained personnel using appropriate standards and procedures.
- Records of calibration should be maintained to provide evidence of the instrument's accuracy over time.
- If an instrument is found to be out of calibration, corrective action should be taken to bring it back into specification.

In summary, calibration is an essential process for ensuring the accuracy and reliability of analytical equipment and apparatus. It should be performed regularly by trained personnel using appropriate standards and procedures. Records of calibration should be maintained to provide evidence of the instrument's accuracy over time. If an instrument is found to be out of calibration, corrective action should be taken to bring it back into specification.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of calcium and magnesium ions.
- EDTA (ethylenediaminetetraacetic acid) is a chelating agent that can bind to these ions, forming a complex.
- In the EDTA method, a solution of EDTA is added to a water sample until all the calcium and magnesium ions have been complexed.
- The amount of EDTA used is then measured, allowing for the calculation of the hardness of the water sample.

- The endpoint of the titration is determined using an indicator such as Eriochrome Black T, which changes color when all the calcium and magnesium ions have been complexed.
- This method is commonly used in laboratories to determine the hardness of water samples.

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the ability of water to neutralize acids. It is determined by measuring the amount of acid required to lower the pH of the water sample to a specific value. The following are the steps involved in the determination of alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the initial pH of the water sample using a pH meter.
3. Add a known amount of acid to the water sample while stirring.
4. Measure the pH of the water sample after each addition of acid.
5. Continue adding acid until the pH of the water sample reaches a specific value.
6. Calculate the amount of acid required to lower the pH of the water sample to the specific value.
7. Use the calculated amount of acid to determine the alkalinity of the water sample.

The alkalinity of the water sample is expressed in terms of milligrams of calcium carbonate (CaCO3) per liter of water. The higher the alkalinity of the water sample, the greater its ability to neutralize acids. Alkalinity is an important parameter in water treatment and environmental monitoring. It is used to assess the buffering capacity of natural waters and to determine the appropriate treatment methods for water purification.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the indicator changes color. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where precise pH measurements are required.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method**: This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: `Surface tension = (weight of liquid raised in the capillary * gravitational acceleration) / (2 * pi * radius of capillary * height of liquid column)`.

2. **Maximum bubble pressure method**: This method involves measuring the maximum pressure required to blow a bubble from the end of a capillary tube immersed in the liquid. The surface tension is then calculated using the formula: `Surface tension = (2 * pressure * radius of capillary) / (4 * radius of bubble)`.

3. **Drop weight method**: This method involves measuring the weight of a droplet of liquid that detaches from the end of a capillary tube. The surface tension is then calculated using the formula: `Surface tension = (weight of droplet * gravitational acceleration) / (pi * radius of capillary^2)`.

4. **Wilhelmy plate method**: This method involves immersing a thin, flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: `Surface tension = (force required to detach plate) / (perimeter of plate)`.

5. **Du Nouy ring method**: This method involves immersing a ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: `Surface tension = (force required to detach ring) / (circumference of ring)`.

These are some of the common methods used to determine the surface tension of a given liquid. Each method has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is commonly referred to as the thickness of a fluid. A viscometer is an instrument used to measure the viscosity of a fluid.

To determine the viscosity of a given liquid using a viscometer, the following steps can be followed:

1. **Preparation:** Clean the viscometer thoroughly and dry it. Make sure the liquid to be tested is at the desired temperature.

2. **Filling the viscometer:** Fill the viscometer with the liquid to be tested up to the marked level.

3. **Measurement:** Start the stopwatch and allow the liquid to flow through the viscometer. Record the time taken for the liquid to flow from the upper to the lower mark.

4. **Calculation:** Use the recorded time and the calibration constant of the viscometer to calculate the viscosity of the liquid.

It is important to note that the viscosity of a liquid can vary with temperature, so it is important to conduct the experiment at a controlled temperature. Additionally, the viscometer should be calibrated before use to ensure accurate results.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. It is commonly used as a primary standard in redox titrations due to its high purity and stability.
3. The strength of a solution of ferrous ammonium sulfate can be determined by titrating it against a standard solution of potassium permanganate or potassium dichromate, using an external indicator such as diphenylamine or starch.
4. The end point of the titration is indicated by a color change of the external indicator, which occurs when all the ferrous ions have been oxidized to ferric ions.
5. The strength of the ferrous ammonium sulfate solution can then be calculated using the volume of the titrant used and the molarity of the standard solution.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is an oxidizing agent that can be used to determine the strength of a substance.

2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator changes color when the reaction is complete, indicating that the titration is finished.
4. The strength of the potassium dichromate can then be calculated using the volume of the titrant and the concentration of the substance being titrated.
5. This method is commonly used in analytical chemistry to determine the strength of various substances.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula Ca(ClO)2. It is commonly used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

The available chlorine in bleaching powder can be determined using the following steps:

1. Weigh a sample of bleaching powder and dissolve it in water.
2. Add excess potassium iodide (KI) to the solution. The iodide ions react with the available chlorine to produce iodine.
3. Titrate the iodine produced with a standard solution of sodium thiosulfate (Na2S2O3) using starch as an indicator. The reaction between iodine and thiosulfate produces iodide ions and sulfate ions.
4. Calculate the amount of available chlorine in the bleaching powder using the volume of thiosulfate solution used in the titration and the concentration of the thiosulfate solution.

This method is known as the iodometric titration method for the determination of available chlorine in bleaching powder. It is a simple and accurate method for measuring the available chlorine content of bleaching powder.

10. Determination of chloride content in water sample

1. Chloride content in water can be determined using a titration method called the Mohr method.
2. The Mohr method involves titrating the water sample with a solution of silver nitrate (AgNO3) using potassium chromate (K2CrO4) as an indicator.
3. The silver nitrate reacts with the chloride ions in the water sample to form a white precipitate of silver chloride (AgCl).
4. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate (Ag2CrO4).
5. The amount of silver nitrate used in the titration can be used to calculate the concentration of chloride ions in the water sample.

6. The concentration of chloride ions can be expressed in milligrams per liter (mg/L) or parts per million (ppm).
7. The maximum acceptable level of chloride in drinking water is 250 mg/L or 250 ppm according to the World Health Organization (WHO) guidelines.
8. High levels of chloride in water can indicate contamination from sources such as sewage, industrial waste, or road salt.
9. Regular monitoring of chloride levels in water sources is important to ensure the safety and quality of drinking water.
10. Other methods for determining chloride content in water include ion chromatography and inductively coupled plasma mass spectrometry (ICP-MS).

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed**: The first step in the preparation of PF resin is to mix phenol and formaldehyde in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol is typically between 1.5:1 and 2:1.

2. **Formation of hydroxymethylphenols**: The reaction between phenol and formaldehyde results in the formation of hydroxymethylphenols, which are intermediate compounds in the synthesis of PF resin.

3. **Polymerization**: The hydroxymethylphenols undergo further reaction to form a linear polymer, known as novolac. The addition of more formaldehyde and heat causes the novolac to crosslink and form a three-dimensional network, resulting in the formation of the final PF resin.

4. **Curing**: The final step in the preparation of PF resin is curing, which involves heating the resin to a high temperature to complete the crosslinking process and form a hard, infusible material.

PF resins are widely used in the production of laminates, adhesives, and molded objects due to their high strength, heat resistance, and chemical resistance. They are also used as binders in the production of abrasive products and as insulation materials in the electrical and electronics industries.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin is a type of thermosetting polymer that is widely used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. **Reacting urea and formaldehyde:** The first step in the preparation of UF resin is the reaction of urea and formaldehyde in an aqueous solution.

The molar ratio of urea to formaldehyde is typically between 1:1.5 and 1:2.5. The reaction is carried out at a pH of 7-8 and a temperature of 85-95°C.

2. **Condensation:** The next step is the condensation of the reaction mixture. This is achieved by heating the mixture to 100-105°C and adjusting the pH to 4.5-5.5. The condensation reaction results in the formation of low molecular weight prepolymers.

3. **Methylation:** The prepolymers are then methylated by adding methanol to the reaction mixture. The methylation reaction is carried out at a temperature of 65-75°C and a pH of 8-9.

4. **Concentration:** The reaction mixture is then concentrated by evaporating the water. This is typically done under reduced pressure at a temperature of 50-60°C.

5. **Curing:** The final step in the preparation of UF resin is curing. This is achieved by heating the resin to a temperature of 130-150°C. The curing process results in the formation of a cross-linked polymer network.

These are the basic steps involved in the preparation of UF resin. The exact conditions and proportions of the reactants may vary depending on the desired properties of the final product. It is important to carefully control the reaction conditions to ensure the formation of a high-quality resin.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and Paracetamol are two different compounds with different preparation methods.

Adipic acid:

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is a white crystalline powder and is mainly used in the production of nylon.

1. Adipic acid can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
2. Another method for the preparation of adipic acid is the carbonylation of butadiene to produce adiponitrile, which is then hydrolyzed to adipic acid.

Paracetamol:

Paracetamol, also known as acetaminophen, is an analgesic and antipyretic drug used to relieve pain and reduce fever.

1. Paracetamol can be prepared by the acetylation of p-aminophenol with acetic anhydride.

2. Another method for the preparation of paracetamol is the nitration of phenol to produce p-nitrophenol, which is then reduced to p-aminophenol and acetylated to produce paracetamol.

14. Determination of Cell Conductance of a solution

1. Cell conductance is a measure of a solution's ability to conduct an electric current.
2. It is determined by measuring the current that flows through a solution between two electrodes of known area and distance apart.
3. The conductance of a solution is directly proportional to the concentration of ions in the solution and the mobility of those ions.
4. The cell constant is determined by measuring the conductance of a solution of known concentration and then using that value to calculate the conductance of an unknown solution.
5. The cell constant is the ratio of the distance between the electrodes to the area of the electrodes.
6. The conductance of a solution can be affected by temperature, so it is important to measure the conductance at a constant temperature.
7. The conductance of a solution can also be affected by the presence of other ions in the solution, so it is important to use a solution that is free of interfering ions when determining the cell conductance.
8. The cell conductance can be used to determine the concentration of an unknown solution by comparing its conductance to the conductance of a solution of known concentration.

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The rate of reaction can be measured by monitoring the change in concentration of one of the reactants or products over time.
3. The rate law for the hydrolysis of esters can be written as: rate = kester, where k is the rate constant, [ester] is the concentration of the ester, and water is the concentration of water.
4. By measuring the rate of reaction at different concentrations of the ester and water, the rate constant k can be determined using the method of initial rates.
5. The method of initial rates involves measuring the initial rate of reaction at different initial concentrations of the reactants.
6. A graph of the initial rate of reaction versus the initial concentration of one of the reactants can be plotted, and the slope of the line can be used to determine the order of the reaction with respect to that reactant.
7. Once the order of the reaction with respect to each reactant is known, the rate constant can be calculated using the rate law equation.
8. The rate constant is an important parameter in understanding the kinetics

of the hydrolysis of esters and can provide valuable information about the mechanism of the reaction.

16. Element detection and identification of functional groups in organic compounds

1. **Element detection** in organic compounds involves the determination of the presence of certain elements such as carbon, hydrogen, nitrogen, sulfur, and halogens.
2. **Functional groups** are specific groups of atoms within molecules that are responsible for the characteristic chemical reactions of those molecules.
3. **Identification of functional groups** in organic compounds can be done through various chemical tests and spectroscopic techniques.
4. Some common functional groups include:
   - **Alcohols** (-OH): can be identified by the Lucas test or by infrared spectroscopy.
   - **Carboxylic acids** (-COOH): can be identified by their reaction with sodium bicarbonate to produce carbon dioxide gas.
   - **Amines** (-NH2): can be identified by the Hinsberg test or by their reaction with nitrous acid to produce nitrogen gas.
   - **Aldehydes** (-CHO): can be identified by the Tollens' test or by their reaction with 2,4-dinitrophenylhydrazine to produce a yellow or orange precipitate.
   - **Ketones** (>C=O): can be identified by their reaction with 2,4-dinitrophenylhydrazine to produce a yellow or orange precipitate.
5. Other techniques for identifying functional groups include mass spectrometry, nuclear magnetic resonance spectroscopy, and ultraviolet-visible spectroscopy.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided above.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions for each experiment.
- Students should also be prepared for the possibility that the instructor may change some of the experiments.
- It is recommended that students communicate with the instructor to clarify any questions or concerns they may have about the experiments.

Course Outcomes:

1. Understanding of the fundamental concepts and principles of the subject.
2. Ability to apply the knowledge and skills acquired in the course to solve problems.
3. Development of critical thinking and analytical skills.
4. Improvement in communication and presentation skills.
5. Enhancement of teamwork and collaboration abilities.
6. Familiarity with the latest developments and trends in the field.
7. Preparation for further studies or professional development in the subject area.

Upon completion of the course the student should be able to:

1. Demonstrate an understanding of the fundamental concepts and principles of the subject.
2. Apply the knowledge and skills acquired in the course to solve problems and complete tasks.
3. Communicate effectively using the terminology and language of the subject.
4. Work collaboratively with others to achieve common goals.
5. Think critically and creatively to analyze and evaluate information.
6. Demonstrate ethical and responsible behavior in the application of the knowledge and skills acquired in the course.
7. Continue to learn and develop their knowledge and skills in the subject through independent study and further education.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is commonly used to design course outcomes and assessments. The taxonomy consists of six levels:

1. **Remembering**: recalling previously learned information.
2. **Understanding**: comprehending the meaning of the information.
3. **Applying**: using the information in a new situation.
4. **Analyzing**: breaking down the information into parts and understanding their relationships.
5. **Evaluating**: making judgments about the value of the information.
6. **Creating**: combining information to form a new whole.

By using Bloom's Taxonomy, educators can design course outcomes that target different levels of cognitive complexity and ensure that students are challenged to develop higher-order thinking skills. This can help students to achieve a deeper understanding of the course material and to apply their knowledge in new and meaningful ways.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness.
- Laser levels project a laser beam to create a straight line on a surface.
- Water levels use the principle that water will always find its own level to determine levelness.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding applications.
- It is important to use a level when installing or building anything that needs to be level or plumb, such as shelves, cabinets, or walls.
- Using a level can help ensure that a project is completed accurately and to a high standard.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are used to measure physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.

3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.

4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Microscopes**: These instruments magnify small objects to make them visible to the naked eye. They are commonly used to study the structure

of cells and tissues.

Each of these instruments has its own unique capabilities and limitations. Understanding how to use these instruments and interpret their results is essential for conducting accurate and reliable analyses.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it is covered by an elastic film, which resists deformation and tends to minimize the surface area of the liquid.

To measure surface tension, several methods can be used, including:

1. The maximum bubble pressure method: This method involves measuring the maximum pressure required to force a gas bubble through a liquid surface. The surface tension is then calculated from the maximum pressure and the dimensions of the tube through which the bubble is forced.

2. The drop weight method: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is then calculated from the weight of the droplet and the dimensions of the tube.

3. The Wilhelmy plate method: This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated from the force and the dimensions of the plate.

These are just a few of the methods that can be used to measure surface tension. Each method has its own advantages and limitations, and the choice of method will depend on the specific requirements of the experiment.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

- **Viscosity** is a measure of a fluid's resistance to flow. It is determined by the internal friction of the fluid, which is caused by the cohesive forces between the molecules. The viscosity of formation water, w, is a function of pressure, temperature, and dissolved solids. In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.

- **Conductance** is the ability of a solution to conduct an electric current. The more ions present in the solution, the higher the conductivity. High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm).

- **Chloride ions** are approximately 1.9 percent of the mass of seawater. The more chloride ions present in the water, the higher the conductivity. In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm.
- **Iron content** in water can affect its taste, color, and odor. Iron can also stain laundry and plumbing fixtures. Iron is commonly found in groundwater and can be removed through various treatment methods.

K3

- K3 is a term that can refer to several different things depending on the context.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are important parameters to measure in water. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the water's ability to neutralize acids.

Here are some points to consider when measuring the hardness and alkalinity of water:

1. Hardness can be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs. The amount of solution required to reach the color change indicates the hardness of the water.
2. Alkalinity can be measured using a similar titration method, where a solution of known concentration is added to a water sample until a pH change occurs. The amount of solution required to reach the pH change indicates the alkalinity of the water.
3. Another method to measure hardness is by using a hardness test strip, which changes color based on the concentration of calcium and magnesium ions in the water.
4. It is important to measure both hardness and alkalinity, as they can affect various processes such as corrosion, scaling, and the effectiveness of water treatment methods.
5. The units of measurement for hardness are typically expressed in milligrams per liter (mg/L) or parts per million (ppm) of calcium carbonate

($CaCO_3$). Alkalinity is typically expressed in mg/L or ppm of $CaCO_3$ as well.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that can be prepared through several methods. Here are some of the methods for the preparation of phenol:

1. **From Haloarenes:** Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .

2. **From Diazonium Salts:** Phenols can also be prepared from diazonium salts.

3. **From Benzene Sulphonic Acid:** At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.

4. **From Cumene:** Phenol can also be prepared by the air oxidation of cumene.

5. **Industrial Preparation Methods:**

    - **From Coal Tar:** Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.
    - **Raschig's Process:** The reaction of benzene, hydrogen chloride and air at 523K.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- **Formaldehyde** is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- **Urea-formaldehyde (UF)** is a nontransparent thermosetting resin or polymer produced from urea and formaldehyde.
- UF is synthesized by the "alkali-acid-alkali" method using formaldehyde and urea as raw materials .
- UF is used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .
- Melamine, polyvinyl alcohol, and adipic acid dihydrazide can be added to UF resin as modifiers for reducing the free formaldehyde.

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific meaning of K3 the user is referring to. Some possible meanings of K3 include:

1. K3 surface: In mathematics, a K3 surface is a complex algebraic surface with trivial canonical bundle.
2. K3 (mountain): K3 is the third highest mountain in the Karakoram range, located in the Gilgit-Baltistan region of Pakistan.
3. K3 (band): K3 is a Belgian-Dutch girl group with a Dutch repertoire, primarily enjoyed by pre-adolescent children.
4. K3 visa: The K3 visa is a nonimmigrant visa for the foreign-citizen spouse of a United States citizen.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by k, is a measure of the speed at which a chemical reaction occurs. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the order of the reaction with respect to each reactant by performing experiments in which the concentration of one reactant is varied while the others are kept constant.
2. Write the rate law for the reaction using the determined orders of reaction.
3. Measure the initial rate of the reaction for different sets of initial concentrations of the reactants.
4. Substitute the measured initial rates and initial concentrations into the rate law to obtain a set of simultaneous equations.
5. Solve the simultaneous equations to obtain the value of the rate constant, k.

It is important to note that the rate constant is temperature-dependent and its value may change with changes in temperature. The relationship between the rate constant and temperature is given by the Arrhenius equation.

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ENGINEERING PHYSICS LAB KCS

Engineering Physics Lab KCS is a laboratory course that is designed to provide students with hands-on experience in the application of physics principles to engineering problems. The course typically covers topics such as:

1. Mechanics: Students learn about the fundamental principles of mechanics, including Newton's laws of motion, conservation of energy, and conservation of momentum.
2. Electricity and Magnetism: Students learn about the fundamental principles of electricity and magnetism, including Coulomb's law, Gauss's law, and Faraday's law of induction.
3. Optics: Students learn about the fundamental principles of optics, including reflection, refraction, and diffraction.
4. Thermodynamics: Students learn about the fundamental principles of thermodynamics, including the first and second laws of thermodynamics, and the concept of entropy.
5. Modern Physics: Students learn about the fundamental principles of modern physics, including the theory of relativity, quantum mechanics, and nuclear physics.

The course is designed to provide students with a strong foundation in the fundamental principles of physics, and to prepare them for further study in engineering and related fields. The laboratory component of the course provides students with the opportunity to apply the principles they have learned in a practical setting, and to develop their skills in experimental design, data analysis, and problem-solving.

Course Objectives:

- To provide students with a comprehensive understanding of the subject matter.
- To develop critical thinking and analytical skills.
- To enhance problem-solving abilities.
- To encourage independent learning and research.
- To prepare students for further studies or careers in the field.
- To promote effective communication and collaboration.
- To foster a sense of social responsibility and ethical behavior.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in various fields such as chemistry, biology, medicine, and environmental science. Here are some fundamental concepts of analytical instruments:

1. **Accuracy and Precision**: Accuracy refers to how close a measurement is to the true value, while precision refers to how close repeated measurements are to each other. Analytical instruments must be both accurate and precise to provide reliable results.

2. **Sensitivity**: Sensitivity refers to the ability of an instrument to detect small changes in the property being measured. A highly sensitive instrument can detect even minute changes, making it useful for detecting trace amounts of substances.

3. **Selectivity**: Selectivity refers to the ability of an instrument to distinguish between different substances. A highly selective instrument can accurately identify and measure specific substances, even in the presence of other substances.

4. **Calibration**: Calibration is the process of adjusting an instrument to ensure that it provides accurate and precise measurements. This is typically done by comparing the instrument's readings to those of a known standard.

5. **Signal-to-Noise Ratio**: The signal-to-noise ratio (SNR) is a measure of the quality of a measurement. It compares the strength of the signal being measured to the background noise. A high SNR indicates a clear and reliable measurement, while a low SNR indicates that the signal is difficult to distinguish from the noise.

These are some of the fundamental concepts of analytical instruments that students should understand. By understanding these concepts, students can better appreciate the capabilities and limitations of analytical instruments and use them effectively in their work.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content:** Chloride content refers to the amount of chloride ions present in a water sample. Chloride ions are negatively charged particles that can be found in many types of water, including seawater, groundwater, and surface water. The presence of chloride ions in water can indicate contamination from sources such as sewage, industrial waste, or road salt.

2. **Hardness:** Water hardness refers to the amount of dissolved minerals, primarily calcium and magnesium, present in a water sample. Hard water can cause problems such as scaling in pipes and appliances, and can interfere with the effectiveness of soaps and detergents. Water hardness

is typically measured in grains per gallon (gpg) or milligrams per liter (mg/L).

3. **Alkalinity:** Alkalinity refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, or its ability to resist changes in pH. Alkalinity is important for maintaining a stable pH in aquatic ecosystems, and can be affected by factors such as pollution and acid rain. Alkalinity is typically measured in milligrams per liter (mg/L) as calcium carbonate (CaCO3).

These parameters are important for understanding the quality of water and can be used to assess its suitability for various uses, including drinking, irrigation, and industrial processes. Analyzing the chloride content, hardness, and alkalinity of water can provide valuable information for managing and protecting water resources.

Water

Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states. It is one of the most plentiful and essential of compounds. A tasteless and odourless liquid at room temperature, it has the important ability to dissolve many other substances.

Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen. Water molecules cling to each other because of a force called hydrogen bonding. It's the reason why water can do amazing things. Water is a shape-shifter. It exists in three states on Earth: liquid, gas, and solid.

Water is the liquid that makes life on Earth possible. As water cycles from the air to the land to the sea and back again, water shapes our planet — and nearly every aspect of our lives. All living things, from tiny cyanobacteria to giant blue whales, need water to survive.

Water is an inorganic compound with the chemical formula H2O. It is a transparent, tasteless, odorless, and nearly colorless chemical substance, and it is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).

Measure of pH, Surface Tension and Viscosity

1. **pH:** pH is a measure of the acidity or basicity of an aqueous solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 indicates an acidic solution, while a pH greater than 7 indicates a basic solution.

2. **Surface Tension:** Surface tension is the property of a liquid that allows

it to resist an external force. It is caused by the cohesive forces between the molecules of the liquid. Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).

3. **Viscosity:** Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity is typically measured in units of pascal-seconds ($Pa \cdot s$) or poise (P). A fluid with a high viscosity, such as honey, flows more slowly than a fluid with a low viscosity, such as water.

These three properties are important in various fields, including chemistry, physics, and engineering. Understanding how to measure and manipulate these properties can be useful in a variety of applications.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to flow and take the shape of its container.
- The particles in a liquid are close together and are constantly moving and sliding past each other, allowing the liquid to flow.
- Liquids have a definite volume, but their shape is determined by the shape of their container.
- The surface of a liquid is typically level, due to the force of gravity pulling the liquid particles downward.
- Liquids can be poured and spilled, and they can also be compressed to some extent.
- The temperature at which a substance changes from a solid to a liquid is called its melting point, and the temperature at which it changes from a liquid to a gas is called its boiling point.
- Some common examples of liquids include water, milk, and oil.

Preparation of Different Resins

Resins can be derived from various sources such as polymers, monomers, oils, fatty acids, and natural sources . They have a wide range of applications in industries such as paints and coatings, personal care and home care, pharmaceuticals, rubber, plastics, electrical and jewelry articles, decorative items, wood coatings, etc .

There are two categories of resinous products: natural resins and prepared resins . So far, no general method has been suggested or proposed for the preparation of resins . However, by manipulating the structure of the linker, resins ranging from extremely acid labile to base labile can be prepared . Using linkers additionally allows the preparation of resins with special applications such as DHP resin utilized as a solid-phase support for alcohols or Weinreb resin utilized for preparing aldehydes and ketones .

There are several types of resins, including but not limited to, epoxy resin, UV resin, polyester resin, and polyurethane resin . The majority of resins are made up of two elements, namely the base resin and a catalyst (which is a hardener). Combining the two elements results in a chemical reaction that allows the resin to set .

Synthesis of Adipic Acid

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is a white crystalline powder that is used primarily as a precursor to the production of nylon. Here are some key points to understand about the synthesis of adipic acid:

1. Adipic acid can be synthesized from a variety of starting materials, including cyclohexanol, cyclohexanone, and benzene.
2. One common method for synthesizing adipic acid is through the oxidation of cyclohexanol or cyclohexanone using nitric acid. This process is known as nitric acid oxidation.
3. Another method for synthesizing adipic acid is through the oxidation of benzene using a cobalt or manganese catalyst. This process is known as the benzene oxidation process.
4. Adipic acid can also be synthesized through the hydrocarboxylation of butadiene, which involves reacting butadiene with carbon monoxide and water in the presence of a catalyst.
5. The choice of synthesis method for adipic acid depends on various factors, including the availability and cost of starting materials, the desired purity of the final product, and environmental considerations.

Acid and Paracetamol by Conventional and Green Route

- Paracetamol, also known as acetaminophen, is a common pain reliever and fever reducer.
- There are several methods for the synthesis of paracetamol, including conventional and green routes.
- Conventional methods for the synthesis of paracetamol involve the use of harsh chemicals and can generate unwanted byproducts.
- Green chemistry principles aim to reduce the environmental impact of chemical processes by minimizing waste, reducing the use of hazardous substances, and increasing energy efficiency.
- An alternative route for the synthesis of paracetamol using green chemistry principles involves solvent-free synthesis combined with microwave-energy

irradiation.

- In 2014, Joncour et al., established a green approach involving direct amidation of hydroquinone to obtain paracetamol with maximum conversion. This approach eliminates the problems associated with earlier techniques, such as the extra step involved, unwanted ortho-isomerization, and generation of the sulfate salt.

List of Experiments

An experiment is simply the test of a hypothesis. A hypothesis, in turn, is a proposed relationship or explanation of phenomena . There are three key types of experiments: controlled experiments, field experiments, and natural experiments .

Some famous experiments include: - Harry Harlow's experiments with baby monkeys and wire and cloth surrogate mothers (1957–1974) - Stanley Milgram's experiments on human obedience (1963) - Walter Mischel's marshmallow experiment showing the importance to life outcomes of the ability to delay gratification (beginning late 1960s)

There are also many science experiments that can be done at home, such as exploring kitchen chemistry or building DIY mini drones . You can find a list of hundreds of fun and easy science experiments to try at home or use for science fair project ideas .

Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing it with a reference standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

1. Calibration is essential for analytical equipment and apparatus to ensure that the measurements obtained are accurate and reliable.
2. The calibration process involves comparing the readings obtained from the instrument with those obtained from a reference standard.
3. The reference standard used for calibration should be traceable to an internationally recognized standard.
4. Calibration should be performed at regular intervals to ensure that the instrument remains accurate over time.
5. The frequency of calibration depends on factors such as the type of instrument, its usage, and the environment in which it is used.
6. Calibration records should be maintained to provide evidence of the instrument's accuracy and reliability.

7. In some cases, calibration may involve adjusting the instrument to bring its readings into alignment with the reference standard.
8. Proper calibration of analytical equipment and apparatus is essential for ensuring the quality and reliability of analytical results.

Determination of Hardness of Water Sample by EDTA Method

The hardness of water is due to the presence of calcium and magnesium salts in water. The determination of hardness of water by EDTA method is one of the three main methods for determining the hardness of water. EDTA stands for Ethylenediaminetetraacetic acid. This EDTA reagent can form EDTA-metal complexes by reacting with metal ions except for alkali metal ions.

Procedure for Calculation of Hardness of Water by EDTA Titration

1. Take a sample volume of 20ml (V ml).
2. Dilute 20ml of the sample in an Erlenmeyer flask to 40ml by adding 20ml of distilled water.
3. Add 1 mL of ammonia buffer to bring the pH to 10 ± 0.1.
4. Add 1 or 2 drops of the indicator solution.
5. Add EDTA titrant to the sample with vigorous shaking until the wine red color just turns blue.

Calculation of Hardness

Hardness (EDTA), as mg/L = $\times$ $\times 1{,}000$ where A = mL of EDTA titrant used, T = Titer of EDTA titrant, mg CaCO3 per mL of EDTA titrant, and S = mL of sample volume. Report as "Total Hardness (EDTA) = ______ mg/L as CaCO3" or as "Calcium Hardness (EDTA) = ______ mg/L as CaCO3".

Determination of Alkalinity of Water Sample

1. Alkalinity is a measure of the ability of water to neutralize acids. It is determined by the presence of ions such as bicarbonate, carbonate, and hydroxide.
2. The alkalinity of a water sample can be determined by titration with a standard acid solution, usually sulfuric acid or hydrochloric acid.
3. The endpoint of the titration is determined by a color change of an indicator, such as phenolphthalein or methyl orange.
4. The amount of acid required to reach the endpoint is used to calculate the alkalinity of the water sample.

5. Alkalinity is typically reported in milligrams per liter (mg/L) as calcium carbonate ($CaCO3$).
6. Alkalinity is an important parameter in water treatment and environmental monitoring, as it affects the pH and buffering capacity of water.
7. High alkalinity can indicate pollution from sources such as agricultural runoff or sewage, while low alkalinity can indicate acid rain or other acidic inputs.

Determination of pH by Titrimetric Method

1. Titration is a laboratory technique used to determine the concentration of an unknown solution by reacting it with a solution of known concentration.
2. The titrimetric method can be used to determine the pH of a solution by titrating it with a strong acid or base.
3. The pH is determined by measuring the volume of titrant required to neutralize the solution.
4. The equivalence point is reached when the number of moles of acid equals the number of moles of base.
5. The pH at the equivalence point can be calculated using the Henderson-Hasselbalch equation.
6. Indicators such as phenolphthalein or bromothymol blue can be used to visually determine the endpoint of the titration.
7. The accuracy of the titrimetric method depends on the precision of the volume measurements and the purity of the reagents used.
8. The titrimetric method is a simple and reliable way to determine the pH of a solution. However, it may not be suitable for solutions with very low or very high pH values.

Determination of Surface Tension of Given Liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height of the liquid that rises in a capillary tube due to the surface tension. The surface tension is then calculated using the formula: `Surface tension = (weight of liquid raised in the capillary * gravitational acceleration) / (2 * pi * radius of capillary * height of liquid raised)`

2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble at the end of a submerged tube. The surface tension is then calculated using the formula:

`Surface tension = (2 * pressure * radius of tube) / (height of liquid column above the tube)`

3. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is then calculated using the formula: `Surface tension = (weight of droplet * gravitational acceleration) / (pi * diameter of tube)`
4. **Wilhelmy plate method:** This method involves measuring the force required to detach a thin plate from the surface of the liquid. The surface tension is then calculated using the formula: `Surface tension = (force required to detach plate) / (perimeter of plate)`
5. **Du Nouy ring method:** This method involves measuring the force required to detach a thin ring from the surface of the liquid. The surface tension is then calculated using the formula: `Surface tension = (force required to detach ring) / (circumference of ring)`

These are some of the common methods used to determine the surface tension of a given liquid. The choice of method may depend on the specific properties of the liquid and the accuracy required in the measurement.

Determination of Viscosity of a given liquid by viscometer

1. Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid.
2. A viscometer is an instrument used to measure the viscosity of a fluid.
3. There are several types of viscometers, including capillary, rotational, and falling ball viscometers.
4. To determine the viscosity of a given liquid using a viscometer, the liquid is placed in the viscometer and the time it takes for the liquid to flow through a specific section of the viscometer is measured.
5. The viscosity of the liquid can then be calculated using the measured flow time and the dimensions of the viscometer.
6. The viscosity of a liquid can also be affected by factors such as temperature and pressure, so these should be controlled during the experiment.
7. The results of the experiment can be used to compare the viscosities of different liquids or to determine the effect of temperature or pressure on the viscosity of a given liquid.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is (NH4)2Fe(SO4)2.6H2O.
3. It is commonly used as a primary standard in redox titrations due to its high purity and stability.
4. To determine the strength of Ferrous ammonium sulfate, a redox titration can be performed using an external indicator.
5. An external indicator is a substance that changes color at the endpoint of the titration, indicating that the reaction is complete.
6. Common external indicators used in redox titrations include diphenylamine, starch, and potassium ferricyanide.
7. The choice of external indicator depends on the specific redox reaction being studied.
8. The procedure for determining the strength of Ferrous ammonium sulfate using an external indicator involves preparing a standard solution of the substance, titrating it against a known solution, and observing the color change of the external indicator to determine the endpoint of the titration.
9. The strength of the Ferrous ammonium sulfate solution can then be calculated using the volume and concentration of the titrant and the balanced chemical equation for the reaction.

Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is potassium chromate.
4. The reaction between potassium dichromate and potassium chromate produces a red-brown color at the end point.
5. The strength of the potassium dichromate solution can be calculated using the volume of the solution used and the concentration of the solution being titrated.

Determination of Available Chlorine in Bleaching Powder

1. The available chlorine present in a bleaching powder sample is determined iodometrically by treating its solution with an excess of potassium iodide solution in an acidic medium.

2. The liberated iodine (I_2) is treated with sodium thiosulphate ($Na_2S_2O_3$) solution using freshly prepared starch solution as an indicator to be added near the endpoint.
3. The chlorine present in the bleaching powder gets reduced with time.
4. The chlorine that is liberated is called available chlorine, and the calculation of available chlorine is called iodometry.
5. Available chlorine is determined by adding a measured amount of sample solution to an acidified solution of potassium iodide, liberating an equivalent amount of iodine. This free iodine is determined by titration with standardized sodium thiosulfate solution, using a starch indicator endpoint.

Determination of Chloride Content in Water Sample

1. Chloride is one of the major anions found in water and wastewater.
2. The chloride content of water can be determined by titration with silver nitrate solution using potassium chromate as an indicator.
3. The reaction between silver nitrate and chloride ions produces a white precipitate of silver chloride.
4. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate.
5. The concentration of chloride ions in the water sample can be calculated from the volume of silver nitrate solution used in the titration.
6. The presence of other anions such as bromide and iodide can interfere with the determination of chloride content.
7. To overcome this interference, the water sample can be treated with nitric acid to convert these anions to their respective acids, which do not interfere with the titration.
8. The determination of chloride content in water is important for monitoring the quality of drinking water and wastewater.
9. High levels of chloride in water can indicate contamination from sources such as industrial discharges or road salt.
10. The acceptable level of chloride in drinking water is typically less than 250 mg/L.

Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol or substituted phenol with formaldehyde. PF resins are formed by a step-growth polymerization reaction that can be either acid- or base-catalyzed .

The preparation of PF resin involves mixing phenol, formaldehyde, water, and a catalyst in the desired amount, depending on the resin to be formed, and then heating the mixture. The first part of the reaction, at around 70 °C, forms a thick reddish-brown tacky material, which is rich in hydroxymethyl and benzylic ether groups.

Since formaldehyde exists predominantly in solution as a dynamic equilibrium of methylene glycol oligomers, the concentration of the reactive form of formaldehyde depends on temperature and pH.

PF resins were the first completely synthetic polymers to be commercialized and were used as the basis for Bakelite .

Preparation of Urea formaldehyde (UF) resin

Urea-formaldehyde (UF) resin, also known as urea-methanal, is a non-transparent thermosetting resin or plastic. It is made from urea and formaldehyde when they are heated in the presence of a mild base such as ammonia or pyridine.

The most common method of preparation for commercial UF resin is the addition of a second amount of urea during the reaction. The ratio of urea to formaldehyde is between 1:2 and 1:2.2 and therefore methylolation can take place at in a short amount of time at temperatures between 90 and 95 °C, with a mixture being maintained under reflux.

One way to prepare UF resin under acidic conditions is to mix 5 g of urea with 6 ml of formalin in a test tube, and shake the test tube until the urea has dissolved. Adjust the pH of the solution to 4 by addition of 4 drops of 0.5N H2SO4, and observe the time required for precipitation to occur.

Another way to prepare a urea-formaldehyde adhesive is to charge sixty grams (1 mole) of urea and 137 g (1.7 mole) formalin into a 500-mL reaction kettle equipped with a mechanical stirrer and a reflux condenser. The pH of the mixture is adjusted with 2M NaOH to between 7 and 8 as determined by universal indicator paper.

The synthesis of all UF resins can be a typical three-step procedure (alkaline-acidic-alkaline). First, formaldehyde (F=250 g) is poured into a three-necked flask and adjusted to pH 7.5 to pH 8.0 with 20% sodium hydroxide solution. The first amount of urea (U 1 =94.4 g) is added to give the F/U molar ratio of 2.0.

Urea is manufactured from carbon dioxide and ammonia at a temperature of 135–200 °C and at a pressure of 70–230 atmospheres. Formaldehyde is manufactured by the oxidation of methanol which can be produced from the reaction of carbon dioxide with hydrogen or can be derived from petroleum.

Preparation of Adipic acid / Paracetamol

Adipic acid

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is the most important dicarboxylic acid from an industrial perspective, with about 2.5 billion kilograms of this white crystalline powder being produced annually, mainly as a precursor for the production of nylon .

Preparation

Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil. The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .

Paracetamol

Paracetamol is an effective antipyretic and analgesic .

Preparation

Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulphuric acid as a catalyst .

Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the current flowing between the electrodes.
4. The cell is filled with the solution to be tested and the conductivity is measured.
5. The cell conductance is calculated by dividing the measured conductivity by the cell constant, which is determined by the geometry of the cell and the distance between the electrodes.
6. The cell constant is usually determined by calibrating the cell with a solution of known conductivity.
7. The cell conductance is affected by the concentration and mobility of the ions in the solution, as well as the temperature of the solution.
8. The cell conductance can be used to determine the concentration of ions in the solution, as well as to monitor changes in the solution, such as the progress of a chemical reaction or the addition of a new solute.
9. The cell conductance is an important parameter in many fields, including chemistry, biology, and environmental science.

Determination of Rate constant of hydrolysis of esters

1. The hydrolysis reaction of an ester in pure water is a slow reaction.
2. When a mineral acid like hydrochloric acid is added, the rate of the reaction is enhanced since the H+ ions from the mineral acid act as the catalyst.
3. The hydrolysis of esters is catalyzed by either an acid or a base.
4. Acidic hydrolysis is simply the reverse of esterification.
5. The ester is heated with a large excess of water containing a strong-acid catalyst.
6. Like esterification, the reaction is reversible and does not go to completion.
7. The rate constant for the hydrolysis reaction of an ester by dilute acid is 0.6931× 10 −3 s −1.
8. The rate constant after calculation from the graphs was approximately 0.003min-1 cm -3 for the 1ml and 2ml of ethyl acetate.
9. When ethyl ethanoate is heated under reflux with a dilute acid such as dilute hydrochloric acid or dilute sulfuric acid, the ester reacts with the water present to produce ethanoic acid and ethanol.

Element detection and identification of functional groups in organic compounds

1. **Element detection** refers to the process of identifying the presence of specific elements in a given compound.
2. **Functional groups** are specific groups of atoms within a molecule that are responsible for the characteristic chemical reactions of that molecule.
3. There are several methods for detecting elements and identifying functional groups in organic compounds, including:
    - **Combustion analysis**: This method involves burning a sample of the compound and analyzing the resulting products to determine the presence of elements such as carbon, hydrogen, and nitrogen.
    - **Infrared spectroscopy**: This technique uses infrared radiation to identify the presence of specific functional groups within a molecule.
    - **Mass spectrometry**: This method involves ionizing a sample of the compound and analyzing the resulting fragments to determine the presence of specific elements and functional groups.
    - **Nuclear magnetic resonance spectroscopy**: This technique uses the interaction of nuclear spins with a magnetic field to identify the presence of specific functional groups within a molecule.
4. These methods can be used in combination to provide a comprehensive analysis of the elements and functional groups present in a given organic compound.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose 10 experiments from the list provided above.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the experiments chosen by the instructor.
- Preparing for these experiments can help students perform well in exams and gain a deeper understanding of the subject matter.

Course Outcomes

Course outcomes are statements that describe the knowledge, skills, and abilities that students are expected to have acquired by the end of a course. These outcomes serve as a guide for instructors in designing course content and assessments, and for students in understanding the goals of the course. Some key points to consider when developing course outcomes include:

1. Course outcomes should be specific, measurable, and achievable within the scope of the course.
2. Outcomes should be aligned with the overall goals of the program or curriculum.
3. Outcomes should be written in clear, concise language that is easily understood by students.
4. Outcomes should be reviewed and updated regularly to ensure they remain relevant and accurate.

By clearly defining course outcomes, instructors can ensure that their course content and assessments are aligned with the goals of the course, and students can better understand what is expected of them and how they will be evaluated. This can help to improve the overall effectiveness of the course and enhance student learning.

Upon completion of the course the student should be able to:

- Demonstrate a comprehensive understanding of the course material.
- Apply the knowledge and skills acquired during the course to solve problems and complete tasks.

- Communicate effectively, both orally and in writing, using the terminology and concepts of the subject.
- Work collaboratively with others to achieve common goals.
- Think critically and analytically, evaluating information and arguments to make informed decisions.
- Demonstrate ethical and professional behavior in all aspects of their work.
- Continue to learn and develop their knowledge and skills in the subject area.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is commonly used to design course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering**: The ability to recall or recognize previously learned information.
2. **Understanding**: The ability to comprehend the meaning of material.
3. **Applying**: The ability to use learned material in new and concrete situations.
4. **Analyzing**: The ability to break down material into its component parts and understand their relationships.
5. **Evaluating**: The ability to make judgments about the value of material for a given purpose.
6. **Creating**: The ability to put parts together to form a new whole.

Course outcomes can be written using Bloom's Taxonomy to ensure that they are specific, measurable, and aligned with the desired level of thinking. For example, a course outcome for a history course might be: "Students will be able to analyze primary source documents to construct an argument about a historical event." This outcome aligns with the "Analyzing" level of Bloom's Taxonomy and specifies the knowledge and skills that students should have acquired by the end of the course.

Level

- A level is a tool used to determine if a surface is horizontal or vertical.
- It consists of a small glass tube filled with liquid, with an air bubble inside.

- When the level is placed on a surface, the bubble will move to the highest point of the tube.
- If the bubble is centered between two marks on the tube, the surface is level.
- Levels come in various sizes and shapes, including torpedo, line, and laser levels.
- They are commonly used in construction, carpentry, and surveying.
- A level can also refer to a relative position or rank in a hierarchy, such as a level of authority or a level of difficulty.
- In video games, a level is a stage or section of the game that the player must complete to progress.
- In education, a level can refer to a stage of proficiency or advancement, such as a grade level or a level of language proficiency.
- In physics, a level can refer to a specific energy state of an atom or molecule.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are used to measure physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including research, quality control, and medical diagnostics. Some common types of analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.

3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.

4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Microscopes**: These instruments magnify small objects or samples. They are commonly used to observe the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations, and the choice of instrument depends on the specific needs of the analysis. Understanding the use of different analytical instruments is essential for selecting the appropriate instrument for a given task and for interpreting the results obtained from the instrument.

CO_2: Measuring Molecular/System Properties such as Surface Tension

Carbon dioxide (CO_2) is a colorless, odorless gas that is naturally present in the Earth's atmosphere. It is an important greenhouse gas that helps regulate the temperature of the planet. However, human activities such as burning fossil fuels and deforestation have increased the concentration of CO_2 in the atmosphere, leading to global warming and climate change.

Surface tension is a property of liquids that arises from the cohesive forces between molecules at the surface of a liquid. It is the force that allows insects to walk on water and causes water droplets to form beads on a surface.

To measure the surface tension of CO_2, several methods can be used, including:

1. The maximum bubble pressure method: This method involves measuring the pressure required to force a gas bubble through a liquid. The surface tension is calculated from the maximum pressure recorded.

2. The drop weight method: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is calculated from the weight of the droplet.

3. The Wilhelmy plate method: This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it. The surface tension is calculated from the force measured.

These are some of the methods that can be used to measure the surface tension of CO_2. It is important to note that the surface tension of CO_2 can vary depending on factors such as temperature and pressure. Therefore, it is important to control these variables when conducting measurements.

Viscosity, Conductance of Solution, Chloride and Iron Content in Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm).
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm.

Chloride Content in Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.
- The more chloride ions present in the water, the higher the conductivity.

Iron Content in Water

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3, a mountain in the Karakoram range in Pakistan, also known as Broad Peak.
2. K3, a Belgian-Dutch girl group.
3. K3, a type of visa for spouses of U.S. citizens to enter the United States while waiting for their immigrant visa to be processed.
4. K3, a type of South Korean high-speed train.
5. K3, a type of tax form used in the United States for reporting income from partnerships, S corporations, and trusts.

CO-3 Measure the hardness and alkalinity of the water

Hardness and alkalinity are two important parameters of water quality. They are related to the concentration of minerals in the water, particularly calcium and magnesium.

- **Hardness** is the measurement of divalent cations (+2 ions) in the water and is expressed as milligrams per liter (mg/L) or parts per million (ppm) of calcium carbonate (CaCO3).
- **Alkalinity** refers to the sum of all bases that buffer against pH swings, with carbonate being the most frequent and important base. Carbonate

hardness, or KH, only measures the concentration of carbonate and bicarbonate bases.

There are several methods to measure the hardness and alkalinity of water. One common method to measure hardness is by using a titration with ethylenediaminetetraacetic acid (EDTA). At a pH of 10, calcium and magnesium ions form colorless, water-soluble complexes with EDTA.

To measure alkalinity, one common method used by the U.S. Geological Survey (USGS) is to take a water sample and add acid to it while checking the pH of the water as the acid is added. An initial pH reading of the water is taken and then small amounts of acid are added in increments, the water is stirred, and the pH is taken.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H+) from the hydroxyl group. There are several methods for the preparation of phenol, including:

1. **From Haloarenes:** Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. **From Benzene Sulphonic Acid:** At 573 − 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
3. **From Cumene:** Phenol can be prepared in large quantities by the air oxidation of cumene.
4. **From Diazonium Salts:** Phenol can also be prepared in small amounts by the hydrolysis of diazonium salts.
5. **From Coal Tar:** Phenol is extracted from the middle oil fraction for commercial purposes at 443 − 503K.
6. **Raschig's Process:** The reaction of benzene, hydrogen chloride and air at 523K produces phenol.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

Formaldehyde

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.

Urea Formaldehyde Resin

- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer.
- It is produced from urea and formaldehyde.
- These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- A stable urea-formaldehyde resin (UF) is synthesized by the "alkali-acid-alkali" method.
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains.

Adipic Acid

- Melamine, polyvinyl alcohol, and adipic acid dihydrazide as modifiers were added to urea-formaldehyde (UF) resin for reducing the free formaldehyde.

Paracetamol

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3 is a Belgian-Dutch girl group with a studio-based line-up.
2. K3 is a mountain in the Karakoram range, also known as Broad Peak.
3. K3 is a visa category for spouses of U.S. citizens.
4. K3 is a type of surface in mathematics, specifically a complex surface with trivial canonical bundle.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k, is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers. The value of k is dependent on the temperature, pressure, and the presence of a catalyst.

There are several methods to estimate the rate constant of a reaction, including:

1. **Graphical methods**: By plotting the concentration of reactants or products against time, the rate constant can be determined from the slope of the line.
2. **Initial rates method**: By measuring the initial rate of the reaction at different initial concentrations of the reactants, the rate constant can be calculated.
3. **Half-life method**: By measuring the time taken for the concentration of a reactant to decrease to half its initial value, the rate constant can be calculated.
4. **Integrated rate laws**: By using the integrated rate laws for zeroth, first, and second-order reactions, the rate constant can be calculated from experimental data.

It is important to note that the units of the rate constant will vary depending on the order of the reaction. For a zeroth-order reaction, the units of k are concentration per unit time, for a first-order reaction, the units of k are per unit time, and for a second-order reaction, the units of k are per unit concentration per unit time.

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- I'm sorry, but I need more information to provide you with the content you are looking for.
- Can you please specify the subject or topic that you want me to write about on page 30 of 40?
- This will help me to provide you with accurate and relevant information.

ENGINEERING PHYSICS LAB

Engineering Physics Lab is a course that is designed to provide students with hands-on experience in the application of physics concepts in engineering. The course typically includes experiments that demonstrate the principles of mechanics, optics, electricity, magnetism, and thermodynamics. Some of the key objectives of the Engineering Physics Lab are:

1. To provide students with an understanding of the experimental techniques used in physics and engineering.
2. To develop students' skills in data analysis and interpretation.
3. To enhance students' ability to work in a team and to communicate their results effectively.
4. To provide students with an opportunity to apply their knowledge of physics to real-world problems.

The experiments in the Engineering Physics Lab are typically designed to be open-ended, allowing students to explore the concepts in depth and to develop their own understanding of the underlying principles. The course also emphasizes the importance of maintaining a detailed laboratory notebook, as this is an essential skill for any scientist or engineer.

Overall, the Engineering Physics Lab is an essential component of any engineering curriculum, as it provides students with the practical skills and knowledge that are necessary to succeed in their future careers.

Course Objectives:

- To provide a comprehensive understanding of the subject matter.
- To develop critical thinking and analytical skills.
- To enhance problem-solving abilities.
- To foster effective communication and collaboration.
- To encourage independent learning and self-motivation.
- To promote the application of theoretical knowledge to real-world scenarios.
- To prepare students for further academic pursuits or professional careers in the field.
- To instill a sense of ethical responsibility and social awareness.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices that are used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, medical diagnosis, and industrial processes.

Some fundamental concepts of analytical instruments include:

- **Accuracy**: The degree to which the measurement result agrees with the true value of the quantity being measured.
- **Precision**: The degree to which repeated measurements of the same quantity agree with each other.
- **Sensitivity**: The smallest change in the quantity being measured that can be detected by the instrument.
- **Resolution**: The smallest difference between two measurements that can be distinguished by the instrument.
- **Calibration**: The process of adjusting the instrument to ensure that its measurements are accurate.
- **Signal-to-noise ratio**: The ratio of the strength of the signal being measured to the background noise.

Understanding these fundamental concepts is essential for students to effectively use and interpret the results of analytical instruments.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water.

- Chloride content in water is measured using a chloride ion selective electrode or by titration with silver nitrate using chromate ions as an indicator.
- Hardness in water is caused by the presence of dissolved calcium and magnesium ions. It can be measured by titration with a chelating agent such as ethylenediaminetetraacetic acid (EDTA) or by using a colorimetric test kit.
- Alkalinity in water is a measure of its capacity to neutralize acids. It is commonly measured by titration with a strong acid such as sulfuric acid or hydrochloric acid, using a pH indicator to determine the endpoint.
- Understanding the levels of these parameters in water is important for various applications, including drinking water treatment, wastewater treatment, and industrial processes.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 is acidic, while a pH greater than 7 is basic. pH can be measured using a pH meter or pH paper.

- **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the liquid molecules. Surface tension can be measured using a number of methods, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.

- **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method.

These properties of liquids are important in many fields, including chemistry, physics, and engineering. Understanding how to measure and interpret these properties can help students better understand the behavior of liquids in various situations.

4. To enable the students to understand about the preparation of different resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to consider when preparing different resins:

1. **Types of resins**: There are several types of resins, including natural resins, synthetic resins, and modified natural resins. Each type of resin has its own unique properties and preparation methods.
2. **Raw materials**: The raw materials used in the preparation of resins vary depending on the type of resin being prepared. For example, natural resins are obtained from plants, while synthetic resins are prepared from chemicals such as formaldehyde and phenol.
3. **Preparation methods**: The methods used to prepare resins also vary depending on the type of resin being prepared. For example, natural resins can be obtained by tapping trees or extracting them from plant material, while synthetic resins are prepared through chemical reactions.
4. **Safety precautions**: When preparing resins, it is important to follow safety precautions to avoid accidents. This includes wearing protective clothing and using proper ventilation.
5. **Quality control**: To ensure that the prepared resins meet the desired specifications, it is important to perform quality control tests. This can include tests for viscosity, hardness, and other properties.

In summary, the preparation of different resins involves understanding the types of resins, the raw materials used, the preparation methods, safety precautions, and quality control. By following these guidelines, students can learn how to prepare different resins effectively.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol by Conventional and Green Route

Adipic acid and paracetamol are two important organic compounds that can be synthesized through both conventional and green routes. The conventional route involves the use of traditional chemical methods, while the green route involves the use of more environmentally friendly methods.

1. **Adipic Acid:** Adipic acid is an important industrial chemical used in the production of nylon. It can be synthesized through the conventional route by the oxidation of cyclohexanol or cyclohexanone with nitric acid. This process produces large amounts of nitrous oxide, a potent greenhouse gas. The green route for the synthesis of adipic acid involves the use of a biocatalyst, such as a lipase enzyme, to catalyze the reaction between cyclohexanol and hydrogen peroxide. This process produces water as the only byproduct, making it a more environmentally friendly option.
2. **Paracetamol:** Paracetamol is a widely used pain reliever and fever reducer. It can be synthesized through the conventional route by the nitration of phenol, followed by the reduction of the nitro group to an amino group, and then acetylation. The green route for the synthesis of paracetamol involves the use of a biocatalyst, such as a nitrilase enzyme, to

catalyze the hydrolysis of 4-aminophenol to paracetamol. This process produces water as the only byproduct, making it a more environmentally friendly option.

In summary, the synthesis of adipic acid and paracetamol can be achieved through both conventional and green routes. The green route, which involves the use of biocatalysts, is a more environmentally friendly option as it produces water as the only byproduct.

LIST OF EXPERIMENTS

- A list experiment is a questionnaire design technique used to mitigate respondent social desirability bias when eliciting information about sensitive topics. With a large enough sample size, list experiments can be used to estimate the proportion of people for whom a sensitive statement is true.
- A list experiment requires that you randomly divide the sample into two groups: the Direct Response Group and the Veiled Response Group. Downsides to list experiments include possible introduction of noise to the data, and possible influence of the RCT treatment on the distributions of responses.
- The official series of experiments, as stated by Jess Winfield, one of the executive producers, are as follows: 0-Series: Jumba's test batch, including many household helpers. 1-Series: Civic disturbances. 2-Series: Technological and scientific. 3-Series: Psychological. 4-Series: Mysterious series of mostly failed experiments.
- Harry Harlow's experiments with baby monkeys and wire and cloth surrogate mothers (1957–1974).
- Stanley Milgram's experiments on human obedience (1963).
- Walter Mischel's marshmallow experiment showing the importance to life outcomes of the ability to delay gratification (beginning late 1960s).
- Science experiments you can do at home! Explore an ever growing list of hundreds of fun and easy science experiments. Have fun trying these experiments at home or use them for science fair project ideas. Explore experiments by category, newest experiments, most popular experiments, easy at home experiments, or ....
- Science Experiments (top 2,000 results) Science Experiments. (top 2,000 results) Fun science experiments to explore everything from kitchen chemistry to DIY mini drones. Easy to set up and perfect for home or school. Browse the collection and see what you want to try first!.

1. Calibration of Analytical Equipment and apparatus

Calibration is the process of verifying the accuracy of an instrument or apparatus by comparing its measurements with a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

- Calibration is typically performed at regular intervals to ensure that the

instrument remains accurate over time.
- The frequency of calibration depends on the instrument and its usage, but it is generally recommended to calibrate at least once a year.
- Calibration can be performed in-house or by an external calibration service.
- The calibration process involves comparing the measurements obtained from the instrument with those obtained from a reference standard.
- The reference standard should be traceable to a national or international standard to ensure the accuracy of the calibration.
- If the instrument is found to be inaccurate, it may need to be adjusted or repaired to bring it back into specification.
- It is important to keep records of calibration to demonstrate that the instrument has been properly maintained and is providing accurate results.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of calcium and magnesium ions.
- EDTA (ethylenediaminetetraacetic acid) is a chelating agent that can bind to these ions and form a complex.
- In the EDTA method, a solution of EDTA is added to a water sample until all the calcium and magnesium ions have formed complexes with the EDTA.
- The amount of EDTA used is then measured, which allows for the determination of the hardness of the water sample.
- The endpoint of the titration is usually indicated by a color change of an indicator such as Eriochrome Black T.
- This method is widely used for the determination of water hardness in both laboratory and field settings.

3. Determination of Alkalinity of water sample

Alkalinity is the measure of the water's ability to neutralize acids. It is an important parameter in water treatment and analysis. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the sample using a pH meter or pH test strips.
3. If the pH is above 7, the water is alkaline.
4. To determine the exact alkalinity, titrate the sample with a standard acid solution, such as sulfuric acid or hydrochloric acid.
5. The endpoint of the titration is reached when the pH of the sample drops to a predetermined value, usually around 4.5.
6. The volume of acid used in the titration is used to calculate the alkalinity of the water sample.

These are the basic steps to determine the alkalinity of a water sample. It is important to follow proper procedures and use accurate equipment to obtain reliable results.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the indicator changes color. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where precise pH measurements are required.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Maximum Bubble Pressure Method**: This method involves measuring the maximum pressure required to blow a bubble of air through a liquid. The surface tension is then calculated using the Laplace equation.

2. **Drop Weight Method**: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is then calculated using the formula for the weight of a droplet.

3. **Capillary Rise Method**: This method involves measuring the height to which a liquid rises in a thin tube due to capillary action. The surface tension is then calculated using the formula for capillary rise.

4. **Wilhelmy Plate Method**: This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula for the force on a flat plate.

5. **Du Nouy Ring Method**: This method involves immersing a thin, circular ring into a liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula for the force on a circular ring.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to carefully calibrate the instruments and follow the experimental procedure to obtain accurate results.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is determined by the internal friction of the fluid, which is caused by the movement of its molecules. A viscometer is an instrument used to measure the viscosity of a fluid.

There are several types of viscometers, including capillary, rotational, and falling ball viscometers. Each type of viscometer operates on a different principle, but all are designed to measure the force required to move an object through a fluid.

To determine the viscosity of a given liquid using a viscometer, the following steps are typically followed:

1. The liquid is poured into the viscometer and allowed to settle.
2. The viscometer is set up according to the manufacturer's instructions.
3. The viscometer is activated, and the time required for the object to move through the liquid is recorded.
4. The viscosity of the liquid is calculated using the recorded time and the known properties of the viscometer.

It is important to note that the viscosity of a liquid can vary with temperature, so it is important to conduct the measurement at a consistent temperature. Additionally, the viscometer should be calibrated regularly to ensure accurate measurements.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is (NH4)2Fe(SO4)2.6H2O.
3. It is commonly used as a primary standard in redox titrations due to its high purity and stability.
4. To determine the strength of ferrous ammonium sulfate, a solution of known concentration is prepared by dissolving a known mass of the salt in a known volume of water.
5. The solution is then titrated against a solution of potassium permanganate, which acts as an oxidizing agent.

6. The end point of the titration is indicated by the appearance of a faint pink color, which is due to the formation of manganese dioxide.
7. The strength of the ferrous ammonium sulfate solution can then be calculated using the balanced chemical equation for the reaction and the volume of potassium permanganate solution used.
8. An external indicator, such as starch, can be used to improve the visibility of the end point.
9. The starch indicator turns blue-black in the presence of iodine, which is produced when potassium iodide is added to the solution.
10. The iodine is then titrated against the ferrous ammonium sulfate solution until the blue-black color disappears, indicating the end point of the titration.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is diphenylamine sulfonic acid.
4. The solution of potassium dichromate is titrated against the solution whose strength is to be determined.
5. The end point of the titration is indicated by the appearance of a blue color.
6. The strength of the solution can be calculated using the formula: Normality of the solution = (Normality of potassium dichromate x Volume of potassium dichromate used) / Volume of the solution used.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula Ca(OCl)2. It is widely used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

The available chlorine in bleaching powder can be determined using the following steps:

1. Weigh a sample of bleaching powder and dissolve it in water to form a solution.
2. Add excess potassium iodide (KI) to the solution. The iodide ions will react with the available chlorine to form iodine.
3. Titrate the iodine formed with a standard solution of sodium thiosulfate (Na2S2O3) using starch as an indicator. The reaction between iodine and thiosulfate is as follows: 2S2O3^2- + I2 -> S4O6^2- + 2I^-

4. From the volume of thiosulfate solution used, the amount of iodine formed can be calculated. From this, the amount of available chlorine in the bleaching powder can be determined.

This method is known as the iodometric titration method for the determination of available chlorine in bleaching powder. It is a simple, accurate, and widely used method in analytical chemistry.

10. Determination of chloride content in water sample

1. Chloride content in water can be determined using a titration method called the Mohr method.
2. The Mohr method involves titrating the water sample with a silver nitrate solution.
3. A chromate indicator is used to signal the end point of the titration.
4. The silver nitrate reacts with the chloride ions in the water sample to form a white precipitate of silver chloride.
5. The chromate ions react with the excess silver ions to form a red precipitate of silver chromate, indicating the end point of the titration.
6. The amount of silver nitrate used in the titration is used to calculate the chloride content in the water sample.
7. The chloride content is usually expressed in milligrams per liter (mg/L) or parts per million (ppm).
8. The Mohr method is a simple and accurate method for determining the chloride content in water.
9. Other methods for determining the chloride content in water include ion chromatography and ion-selective electrodes.
10. The chloride content in water is an important parameter for monitoring water quality, as high levels of chloride can indicate pollution from sources such as road salt or industrial waste.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. Phenol and formaldehyde are mixed in the presence of an acidic or basic catalyst.
2. The reaction is carried out at a temperature of around 80-90°C.
3. The ratio of formaldehyde to phenol is typically between 1.5:1 and 2:1.
4. The reaction mixture is stirred continuously to ensure uniform mixing.
5. The reaction is allowed to proceed until the desired degree of polymerization is achieved.
6. The resulting resin is then cooled and solidified.
7. The solid resin is ground into a fine powder, which can be used as a molding compound or as a bonding agent in the manufacture of various

products.

PF resins are widely used in the production of laminates, plywood, and other wood products, as well as in the manufacture of electrical components, brake linings, and other industrial products. They are known for their high strength, durability, and resistance to heat and chemicals.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin is a type of thermosetting polymer that is commonly used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. The first step in the preparation of UF resin is the reaction between urea and formaldehyde. This reaction takes place in an aqueous solution, with the molar ratio of urea to formaldehyde typically ranging from 1:1.5 to 1:2.5.
2. The reaction is carried out under acidic conditions, with the pH of the solution being maintained between 1 and 3. The temperature of the reaction is typically between 60 and 95°C.
3. The reaction between urea and formaldehyde produces a number of intermediate compounds, including monomethylol urea and dimethylol urea. These intermediates can then react with each other to form a variety of oligomers.
4. The next step in the preparation of UF resin is the addition of a hardening agent, such as ammonia or an amine. This causes the oligomers to crosslink, forming a three-dimensional network.
5. The final step in the preparation of UF resin is the removal of water and other volatile compounds. This is typically achieved through a combination of heating and vacuum distillation.
6. The resulting UF resin can then be used in a variety of applications, including as an adhesive in the production of wood-based panels, as a finish for textiles, and as a molding compound in the production of plastic objects.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and Paracetamol are two different compounds with different preparation methods.

Adipic Acid:

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is a white crystalline powder and is mainly used in the production of nylon.

1. Adipic acid can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.

2. Another method involves the oxidation of cyclohexane with air in the presence of a cobalt catalyst.
3. Adipic acid can also be prepared by the hydrolysis of acrylonitrile.

Paracetamol:

Paracetamol, also known as acetaminophen, is an over-the-counter pain reliever and fever reducer.

1. Paracetamol can be prepared by the nitration of phenol, followed by the reduction of the nitro group to an amine.
2. Another method involves the acetylation of p-aminophenol with acetic anhydride.
3. Paracetamol can also be prepared by the hydrolysis of phenacetin.

14. Determination of Cell Conductance of a solution

1. Cell conductance is a measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the current passing through the solution.
4. The cell constant is determined by measuring the conductivity of a solution of known conductance.
5. The conductance of the solution is then calculated by dividing the measured conductivity by the cell constant.
6. The conductance of the solution is affected by factors such as the concentration and mobility of the ions in the solution, the temperature of the solution, and the presence of other substances in the solution.
7. The conductance of a solution can be used to determine the concentration of ions in the solution, to monitor the progress of chemical reactions, and to measure the purity of water.

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of the hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The reaction between an ester and water to produce a carboxylic acid and an alcohol is known as hydrolysis.
3. The rate of hydrolysis of an ester can be determined by measuring the change in concentration of one of the reactants or products over time.
4. The rate law for the hydrolysis of an ester can be written as: rate = kester, where k is the rate constant, [ester] is the concentration of the ester, and water is the concentration of water.
5. By measuring the initial rate of reaction at different initial concentrations of the ester and water, the rate constant k can be determined using the method of initial rates.

6. The rate constant can also be determined by measuring the change in concentration of one of the reactants or products over time and fitting the data to the integrated rate law for a first-order reaction.
7. The rate constant is an important parameter in understanding the kinetics of the hydrolysis of esters and can provide valuable information about the mechanism of the reaction.

16. Element detection and identification of functional groups in organic compounds

1. **Element detection** refers to the process of identifying the presence of specific elements in a given sample.
2. **Functional groups** are specific groups of atoms within molecules that are responsible for the characteristic chemical reactions of those molecules.
3. The identification of functional groups in organic compounds is important for understanding their chemical properties and reactivity.
4. Common methods for element detection and identification of functional groups in organic compounds include:
   - **Infrared spectroscopy (IR)**: This technique measures the absorption of infrared radiation by the sample, which can provide information about the functional groups present.
   - **Mass spectrometry (MS)**: This technique measures the mass-to-charge ratio of ions produced from the sample, which can provide information about the molecular weight and structure of the compound.
   - **Nuclear magnetic resonance spectroscopy (NMR)**: This technique measures the absorption of radiofrequency radiation by the sample in the presence of a magnetic field, which can provide information about the chemical environment of the atoms in the molecule.
   - **Ultraviolet-visible spectroscopy (UV-Vis)**: This technique measures the absorption of ultraviolet and visible radiation by the sample, which can provide information about the electronic structure of the molecule.
5. These techniques can be used in combination to provide a more complete picture of the composition and structure of the sample.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose 10 experiments from the list provided above.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the experi-

ments chosen by the instructor.
- Students should prepare for the experiments by reviewing the relevant material and asking questions if necessary.
- By completing the experiments, students will gain hands-on experience and a deeper understanding of the subject matter.

Course Outcomes:

- Define and explain the key concepts and principles of the subject.
- Demonstrate an understanding of the subject's theories and their applications.
- Analyze and evaluate information, arguments, and evidence related to the subject.
- Apply the subject's concepts and principles to solve problems and make decisions.
- Communicate effectively in written and oral forms about the subject.
- Work effectively in a team to achieve common goals related to the subject.
- Demonstrate ethical and professional behavior in the context of the subject.
- Engage in continuous learning and professional development related to the subject.

Upon completion of the course the student should be able to:

1. Demonstrate a thorough understanding of the course material.
2. Apply the concepts and theories learned in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.
4. Communicate effectively in both written and oral forms.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior.
7. Utilize technology effectively to enhance learning and problem-solving.
8. Engage in lifelong learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. The taxonomy consists of six levels: remembering, understanding, applying, analyzing, evaluating, and creating. These levels are often used to structure course outcomes and to design assessments that measure student achievement of the outcomes.

Here are some key points to consider when using Bloom's Taxonomy to develop course outcomes:

1. Start by identifying the overall goals and objectives of the course.
2. Use Bloom's Taxonomy to categorize the goals and objectives into different levels of complexity and specificity.
3. Write clear and specific course outcomes that describe what students should be able to do by the end of the course.
4. Align the course outcomes with the course objectives and the levels of Bloom's Taxonomy.
5. Design assessments that measure student achievement of the course outcomes at the appropriate level of Bloom's Taxonomy.

By using Bloom's Taxonomy to develop course outcomes, instructors can ensure that their courses are designed to effectively promote student learning and achievement. This can help students to develop the knowledge, skills, and abilities that are necessary for success in their academic and professional careers.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness.
- Laser levels project a laser beam to create a straight line on a surface.
- Water levels use the principle that water will always find its own level to determine levelness.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding applications.
- It is important to use a level when installing or building anything that needs to be level or plumb, such as shelves, cabinets, or walls.
- Using a level can help ensure that a project is completed accurately and to a high standard.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are devices used to measure, analyze, and quantify the physical and chemical properties of a sample. These instruments are commonly used in scientific research, quality control, and medical diagnosis. Some of the most common analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs**: These instruments separate and analyze the components of a mixture. There are several types of chromatography, including gas chromatography, liquid chromatography, and thin-layer chromatography.

3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.

4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Microscopes**: These instruments magnify small objects to make them visible to the naked eye. They are commonly used to study the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations, and the choice of instrument depends on the specific needs of the analysis. Understanding the use of different analytical instruments is essential for conducting accurate and reliable scientific research.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it is covered by an elastic film. Surface tension is typically measured in units of force per unit length, such as N/m or dyn/cm.

There are several methods for measuring surface tension, including:

1. The maximum bubble pressure method: This method involves measuring the maximum pressure required to blow a bubble through a liquid using a capillary tube.
2. The drop weight method: This method involves measuring the weight of drops of liquid that detach from the end of a thin tube.
3. The Wilhelmy plate method: This method involves measuring the force required to detach a thin, flat plate from the surface of a liquid.
4. The Du Nouy ring method: This method involves measuring the force required to detach a thin, circular ring from the surface of a liquid.

Each of these methods has its own advantages and disadvantages, and the choice of method will depend on the specific properties of the liquid being measured and the accuracy required for the measurement. It is important to carefully calibrate and control the experimental conditions when measuring surface tension to ensure accurate and reliable results.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

- **Viscosity** is a measure of a fluid's resistance to flow. It is affected by factors such as pressure, temperature, and dissolved solids. In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature. Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.
- **Conductance** is the ability of a solution to conduct electricity. The more ions present in the water, the higher the conductivity. High quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while sea water is about 50 mS/cm (or 0.05 S/cm).
- **Chloride ions** are approximately 1.9 percent of the mass of seawater. The more chloride ions present in the water, the higher the conductivity. In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm.
- **Iron content** in water can affect its taste, color, and odor. Iron can also cause staining of laundry and plumbing fixtures. Iron is commonly found in groundwater and can be removed through various treatment methods.

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3, a Belgian-Dutch girl group formed in 1998.
2. K3, a mountain in the Karakoram range, also known as Broad Peak.
3. K3, a type of visa for spouses of U.S. citizens to enter the United States while waiting for their immigrant visa to be processed.
4. K3, a type of high-speed train operated by China Railway High-speed.
5. K3, a type of potassium channel found in the nervous system.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters to measure the quality of water. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the water's ability to neutralize acids.

1. To measure the hardness of water, a titration method can be used. This involves adding a known amount of a chemical called ethylenediaminetetraacetic acid (EDTA) to a water sample. The EDTA binds with the calcium and magnesium ions, and the amount of EDTA used can be used to calculate the hardness of the water.

2. To measure the alkalinity of water, a titration method can also be used. This involves adding a known amount of acid to a water sample until the pH reaches a certain value. The amount of acid used can be used to calculate the alkalinity of the water.

It is important to measure the hardness and alkalinity of water because they can affect the taste, odor, and appearance of the water, as well as its ability to form scale or corrode pipes. These measurements can also be used to determine the appropriate treatment methods for the water.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H+) from the hydroxyl group. There are several methods for the preparation of phenol, including:

1. From Haloarenes: Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .

2. From Benzene Sulphonic Acid: Sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide at 573 – 623K. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.

3. From Cumene: Phenol can be prepared by the air oxidation of cumene.

4. From Diazonium Salts: Small amounts of phenol can be prepared by the hydrolysis of diazonium salts.

5. From Coal Tar: Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.

6. Raschig's Process: The reaction of benzene, hydrogen chloride, and air at 523K produces phenol.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer produced from urea and formaldehyde.
- UF is synthesized by the "alkali-acid-alkali" method .
- UF is used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .

- Melamine, polyvinyl alcohol, and adipic acid dihydrazide can be added to UF resin for reducing the free formaldehyde.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k, is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective orders. The units of the rate constant depend on the order of the reaction.

There are several methods to estimate the rate constant of a reaction, including:

1. **Graphical methods:** These methods involve plotting the concentration of reactants or products against time and determining the slope of the line or curve. The slope is related to the rate constant of the reaction.

2. **Initial rates method:** This method involves measuring the initial rate of the reaction at different initial concentrations of the reactants. The rate constant can be calculated by fitting the data to the rate law equation.

3. **Half-life method:** This method involves measuring the time it takes for the concentration of a reactant to decrease to half of its initial value. The rate constant can be calculated using the half-life equation.

4. **Integrated rate law method:** This method involves measuring the concentration of a reactant at different times and fitting the data to the integrated rate law equation. The rate constant can be calculated from the parameters of the fitted equation.

It is important to note that the rate constant is temperature-dependent and can change with changes in temperature. The Arrhenius equation can be used to calculate the rate constant at different temperatures.

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- I'm sorry, but I need more information to provide you with the content you are looking for.
- Can you please specify the subject or topic that you want me to write about on page 31 of 40?
- Providing more details will help me to generate the content that you need for your studies.

Engineering Mathematics Lab KCS

Engineering Mathematics Lab KCS is a course that focuses on the practical application of mathematical concepts and techniques in engineering. The course is designed to provide students with hands-on experience in using mathematical tools and software to solve engineering problems.

Some of the key topics covered in this course include:

1. Numerical methods for solving equations and systems of equations.
2. Techniques for curve fitting and interpolation.
3. Numerical integration and differentiation.
4. Optimization techniques.
5. Probability and statistics for engineering applications.

The course typically involves a combination of lectures, tutorials, and laboratory sessions, where students have the opportunity to work on practical problems and apply the mathematical concepts and techniques they have learned.

The course is typically taken by students in their second or third year of an engineering program, and is a core requirement for many engineering degrees. It provides students with a solid foundation in the mathematical skills and techniques that are essential for success in engineering.

Course Objectives:

1. To provide students with a comprehensive understanding of the subject matter.
2. To develop critical thinking and problem-solving skills.
3. To encourage independent learning and self-directed study.
4. To prepare students for further academic or professional pursuits in the field.
5. To foster an appreciation for the subject and its relevance to the world around us.
6. To promote effective communication and collaboration skills.
7. To instill a sense of ethical responsibility and social awareness.
8. To provide a foundation for lifelong learning and personal growth.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in scientific research, medical diagnosis, and industrial processes. Here are some fundamental concepts of analytical instruments:

1. **Accuracy**: This refers to how close the measurement is to the true value of the substance being analyzed. High accuracy is essential for reliable results.

2. **Precision**: This refers to how consistent the measurements are when the same substance is analyzed multiple times. High precision is important for reproducibility of results.

3. **Sensitivity**: This refers to the ability of the instrument to detect small changes in the substance being analyzed. High sensitivity is important for detecting low concentrations of substances.

4. **Selectivity**: This refers to the ability of the instrument to distinguish between different substances. High selectivity is important for accurate identification of substances.

5. **Calibration**: This is the process of adjusting the instrument to ensure that it is providing accurate and precise measurements. Regular calibration is essential for maintaining the reliability of the instrument.

6. **Maintenance**: Regular maintenance is important to ensure that the instrument is functioning properly and providing accurate and reliable results.

These are some of the fundamental concepts of analytical instruments that students should understand. Understanding these concepts will enable students to use analytical instruments effectively and accurately in their work.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content:** Chloride is one of the major anions found in water and wastewater. The analysis of chloride content in water is important because high levels of chloride can be harmful to aquatic life and can also affect the taste of drinking water.

2. **Hardness:** Hardness in water is caused by the presence of dissolved calcium and magnesium ions. Hardness is usually expressed in terms of the equivalent quantity of calcium carbonate. The analysis of hardness in water is important because it can affect the performance of water-using appliances and can also cause scaling in pipes and boilers.

3. **Alkalinity:** Alkalinity is a measure of the capacity of water to neutralize acids. It is primarily due to the presence of bicarbonate, carbonate, and hydroxide ions. The analysis of alkalinity in water is important because it can affect the pH of the water and can also influence the effectiveness of water treatment processes.

These are some of the key points to understand about the analysis of chloride content, hardness, and alkalinity in water. It is important for students to have a good understanding of these concepts in order to properly analyze and interpret water quality data.

Water

- Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states.
- It is one of the most plentiful and essential of compounds.
- A tasteless and odourless liquid at room temperature, it has the important ability to dissolve many other substances.
- Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen.
- Water molecules cling to each other because of a force called hydrogen bonding.
- Water is a shape-shifter. It exists in three states on Earth: liquid, gas, and solid.
- Water is the liquid that makes life on Earth possible.
- As water cycles from the air to the land to the sea and back again, water shapes our planet and nearly every aspect of our lives.
- All living things, from tiny cyanobacteria to giant blue whales, need water to survive.
- Water is an inorganic compound with the chemical formula H2O.
- It is a transparent, tasteless, odorless, and nearly colorless chemical substance.
- It is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).

Measure of pH, Surface Tension and Viscosity

1. **pH** is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 indicates an acidic solution, while a pH greater than 7 indicates a basic solution.

2. **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the molecules

of the liquid. Surface tension can be measured using a number of methods, including the maximum bubble pressure method, the drop weight method, and the Wilhelmy plate method.

3. **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method.

These are important concepts in chemistry and physics, and understanding them can help students better understand the behavior of fluids and solutions.

Liquid

- A liquid is a state of matter that has a definite volume but no fixed shape.
- Liquids take the shape of their container and can flow easily.
- The molecules in a liquid are close together and are in constant motion.
- The motion of the molecules in a liquid allows it to change shape and flow.
- Liquids have a higher density than gases but a lower density than solids.
- The temperature at which a substance changes from a solid to a liquid is called its melting point.
- The temperature at which a substance changes from a liquid to a gas is called its boiling point.
- Liquids can be described by their properties such as viscosity, surface tension, and density.
- Viscosity is a measure of a liquid's resistance to flow.
- Surface tension is the property of a liquid that allows it to resist an external force.
- Density is the mass of a substance per unit volume.

Preparation of Different Resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are used in a variety of applications, including adhesives, coatings, and plastics. Here are some points on the preparation of different resins:

1. **Phenolic resins:** Phenolic resins are prepared by the reaction of phenol with formaldehyde in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol, the type of catalyst used, and the reaction conditions determine the properties of the resulting resin.

2. **Epoxy resins:** Epoxy resins are prepared by the reaction of epichlorohydrin with bisphenol A. The resulting resin is then cured with a hardener, such as an amine or anhydride, to form a crosslinked polymer.

3. **Polyester resins:** Polyester resins are prepared by the reaction of a dibasic acid, such as phthalic anhydride, with a diol, such as ethylene

glycol. The resulting resin is then cured with a hardener, such as styrene, to form a crosslinked polymer.

4. **Silicone resins:** Silicone resins are prepared by the hydrolysis and condensation of silanes or siloxanes. The resulting resin can be cured with heat or a catalyst to form a crosslinked polymer.

These are just a few examples of the many different types of resins and their preparation methods. Each type of resin has its own unique properties and applications. It is important for students to understand the preparation methods and properties of different resins in order to select the appropriate resin for a given application.

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other industrial products. Here are some key points to understand about the synthesis of adipic acid:

1. Adipic acid is synthesized from cyclohexene or cyclohexanone through a process called oxidation.
2. The oxidation process involves the use of an oxidizing agent, such as nitric acid, to convert the starting material into adipic acid.
3. The reaction can be carried out in the presence of a catalyst, such as copper or vanadium, to increase the reaction rate and improve the yield of adipic acid.
4. The reaction conditions, such as temperature and pressure, can be adjusted to optimize the synthesis of adipic acid.
5. The resulting adipic acid can be purified through crystallization or other separation techniques to obtain a high-purity product.

Acid and Paracetamol by Conventional and Green Route

- An alternative route for the synthesis of N-acetyl-p-aminophenol (paracetamol, PAR) and acetylsalicylic acid (aspirin, ASA) using principles of green chemistry has been proposed. This involves solvent-free synthesis combined with microwave-energy irradiation.
- In 2014, Joncour et al., established a green approach involving direct amidation of hydroquinone to obtain paracetamol with maximum conversion. This eliminates the problems associated with the earlier techniques such as, the extra step involved, unwanted ortho- isomerization, generation of the sulfate salt.
- Acetaminophen can be synthesized from phenol in three steps. In this synthetic route, the solvent from step two is kept to help maximize atom

economy. The first step is an electrophilic aromatic substitution on phenol with nitric acid to create pnitrophenol.

List of Experiments

1. **The Milgram Experiment:** A series of social psychology experiments conducted by Stanley Milgram in the 1960s, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts that conflicted with their personal conscience.
2. **The Stanford Prison Experiment:** A social psychology experiment conducted by Philip Zimbardo in 1971, which aimed to investigate the psychological effects of perceived power, focusing on the struggle between prisoners and prison officers.
3. **The Asch Conformity Experiment:** A series of studies conducted by Solomon Asch in the 1950s, which demonstrated the power of conformity in groups and showed that people often conform to the opinions of the majority, even when those opinions are clearly wrong.
4. **The Pavlov's Dog Experiment:** A series of experiments conducted by Ivan Pavlov in the 1890s, which demonstrated the principle of classical conditioning, where a neutral stimulus can be used to elicit a conditioned response.
5. **The Marshmallow Test:** A study on delayed gratification conducted by Walter Mischel in the 1960s and 1970s, which showed that children who were able to delay gratification and resist the temptation to eat a marshmallow immediately in favor of a larger reward later on, tended to have better life outcomes in areas such as education and health.

Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of an instrument or apparatus by comparing its measurements with a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

1. Calibration is necessary for all analytical equipment and apparatus, including balances, pipettes, thermometers, and spectrophotometers.
2. The frequency of calibration depends on the instrument and its usage. Some instruments may require daily calibration, while others may only need to be calibrated once a year.
3. Calibration can be performed using certified reference materials, which are materials with known properties that have been verified by an independent organization.
4. Calibration can also be performed using internal standards, which are materials with known properties that are prepared and verified by the

laboratory itself.
5. Calibration records should be maintained to document the calibration process and results.
6. Regular calibration is essential for ensuring the accuracy and reliability of analytical results. It is also a requirement for compliance with quality standards such as ISO 17025.

Determination of Hardness of Water Sample by EDTA Method

The hardness of water is due to the presence of calcium and magnesium salts in water. The EDTA method is one of the three main methods for determining the hardness of water. EDTA stands for Ethylenediaminetetraacetic acid. This reagent can form EDTA-metal complexes by reacting with metal ions, except for alkali metal ions.

In the EDTA method, the water is titrated with ethylene diamine tetraacetic acid using Erichrome Black T (EBT) as an indicator. The procedure for calculating the hardness of water by EDTA titration is as follows:

1. Take a sample volume of 20ml (V ml).
2. Dilute 20ml of the sample in an Erlenmeyer flask to 40ml by adding 20ml of distilled water.
3. Add 1 mL of ammonia buffer to bring the pH to 10 ± 0.1.
4. Add 1 or 2 drops of the indicator solution.
5. Add EDTA titrant to the sample with vigorous shaking until the wine red color just turns blue.

The hardness can then be calculated as follows:

Hardness (EDTA), as mg/L = (A * T * 1000) / S

Where: - A = mL of EDTA titrant used - T = Titer of EDTA titrant, mg CaCO3 per mL of EDTA titrant - S = mL of sample volume

The result can be reported as "Total Hardness (EDTA) = ______ mg/L as CaCO3" or as "Calcium Hardness (EDTA) = ______ mg/L as CaCO3".

Determination of Alkalinity of Water Sample

1. Alkalinity is a measure of the ability of water to neutralize acids. It is determined by the concentration of bicarbonate, carbonate, and hydroxide ions present in the water sample.
2. The alkalinity of a water sample can be determined by titration with a standard acid solution, usually sulfuric acid or hydrochloric acid.
3. The endpoint of the titration is determined by a color change of an indicator, such as phenolphthalein or bromcresol green-methyl red.

4. The volume of acid required to reach the endpoint is used to calculate the alkalinity of the water sample.
5. The result is usually expressed in milligrams per liter (mg/L) of calcium carbonate (CaCO3) equivalent.
6. Alkalinity is an important parameter in water treatment and environmental monitoring, as it affects the pH and buffering capacity of water.

Determination of pH by Titrimetric Method

Titration is a laboratory technique used to determine the concentration of an unknown solution by adding a solution of known concentration until the reaction between the two solutions is complete. The point at which the reaction is complete is called the endpoint, and it is usually indicated by a color change.

The titrimetric method can be used to determine the pH of a solution by using an acid-base titration. In an acid-base titration, an acid or base of known concentration (the titrant) is added to a solution of the substance being analyzed (the analyte) until the reaction is complete.

Here are the steps involved in determining the pH of a solution using the titrimetric method:

1. Choose an appropriate indicator that changes color at the desired pH range.
2. Measure a known volume of the analyte solution into a flask or beaker.
3. Add the titrant solution to the analyte solution while stirring until the endpoint is reached, as indicated by the color change of the indicator.
4. Record the volume of titrant solution used to reach the endpoint.
5. Use the volume and concentration of the titrant solution and the volume of the analyte solution to calculate the concentration of the analyte solution.
6. Use the concentration of the analyte solution and the dissociation constant of the acid or base to calculate the pH of the solution.

This method is commonly used to determine the pH of solutions in laboratories and is a reliable and accurate way to measure the acidity or basicity of a solution. It is important to choose the right indicator and to carefully measure the volumes of the solutions used in order to obtain accurate results.

Determination of Surface Tension of Given Liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface. It is defined as the force acting per unit length perpendicular to the line of separation of the liquid surface from the air.

There are several methods to determine the surface tension of a given liquid. Some of the common methods are:

1. **Capillary rise method**: This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid.
2. **Maximum bubble pressure method**: This method involves measuring the maximum pressure required to blow a bubble of air through a liquid.
3. **Drop weight method**: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube.
4. **Wilhelmy plate method**: This method involves measuring the force required to detach a thin flat plate from the surface of a liquid.
5. **Du Nouy ring method**: This method involves measuring the force required to detach a thin ring from the surface of a liquid.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to follow the standard procedures and calibrate the instruments properly to obtain accurate results.

Determination of Viscosity of a given liquid by viscometer

1. Viscosity is a measure of a fluid's resistance to flow.
2. A viscometer is an instrument used to measure the viscosity of a liquid.
3. There are several types of viscometers, including capillary, falling ball, and rotational viscometers.
4. The method used to determine the viscosity of a liquid using a viscometer depends on the type of viscometer being used.
5. For example, in a capillary viscometer, the time it takes for a liquid to flow through a narrow tube is measured and used to calculate the viscosity of the liquid.
6. In a falling ball viscometer, the time it takes for a ball to fall through a liquid is measured and used to calculate the viscosity of the liquid.
7. In a rotational viscometer, the torque required to rotate a spindle in a liquid is measured and used to calculate the viscosity of the liquid.
8. The viscosity of a liquid can be affected by factors such as temperature and pressure, so it is important to control these variables during the measurement.
9. The viscosity of a liquid is typically reported in units of poise or centipoise.
10. The viscosity of a liquid can provide important information about its physical properties and behavior.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is (NH4)2Fe(SO4)2.6H2O.
3. It is commonly used as a primary standard in redox titrations due to its high purity and stability.
4. The strength of a solution of ferrous ammonium sulfate can be determined by titrating it against a standard solution of potassium permanganate (KMnO4) using an external indicator.
5. The external indicator used in this titration is potassium ferricyanide (K3Fe(CN)6).
6. The reaction between ferrous ammonium sulfate and potassium permanganate is a redox reaction, where potassium permanganate acts as an oxidizing agent and ferrous ammonium sulfate acts as a reducing agent.
7. The end point of the titration is indicated by the disappearance of the blue color of the potassium ferricyanide solution, which indicates that all the ferrous ions have been oxidized to ferric ions.
8. The strength of the ferrous ammonium sulfate solution can then be calculated using the volume of potassium permanganate solution used and the molarity of the potassium permanganate solution.

Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is potassium iodide.
4. The reaction between potassium dichromate and potassium iodide produces iodine, which can be detected by its characteristic color.
5. The end point of the titration is reached when all the iodine has been consumed and the solution turns from a yellow color to a colorless solution.
6. The strength of the potassium dichromate solution can be calculated using the volume of the solution used and the concentration of the iodine solution.

Determination of Available Chlorine in Bleaching Powder

- The available chlorine present in a bleaching powder sample is determined iodometrically by treating its solution with an excess of potassium iodide solution in an acidic medium.

- The liberated iodine (I2) is treated with sodium thiosulphate (Na2S2O3) solution using freshly prepared starch solution as an indicator to be added near the endpoint.
- The chlorine present in the bleaching powder gets reduced with time.
- The chlorine that is liberated is called available chlorine, and the calculation of available chlorine is called iodometry.
- A measured amount of sample solution is added to an acidified solution of potassium iodide, liberating an equivalent amount of iodine. This free iodine is determined by titration with standardized sodium thiosulfate solution, using a starch indicator endpoint.

Determination of Chloride Content in Water Sample

The amount of chloride in water can be determined by several methods, including titration with silver nitrate solution using potassium chromate indicator, also known as Mohr's method. The reaction is quantitative, with AgNO3 reacting with chloride ions in a 1:1 ratio.

The procedure for determining chloride content in water using this method is as follows:

1. Take 50mL of the water sample and dilute it to 100mL.
2. If the sample is highly colored, add 3mL of aluminum hydroxide, shake well, allow to settle, filter, wash, and collect the filtrate.
3. Bring the sample pH to 7-8 by adding acid or alkali.
4. Add 1mL of potassium chromate indicator.
5. Titrate the solution against a standard silver nitrate solution until a reddish-brown precipitate is obtained.

This method can be used to determine the chloride ion concentration of water samples from many sources, such as seawater, stream water, river water, and estuary water. The pH of the sample solutions should be between 6.5 and 10.

Another method for determining the concentration of chloride ions in water is by using a chloride-specific ion electrode and potentiometry. In this method, the potential of an electrochemical cell is measured while little to no current is passed through the sample.

Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol or substituted phenol with formaldehyde. Phenol-formaldehyde resins are formed by a step-growth polymerization reaction that can be either acid or base-catalyzed.

The preparation of Phenol Formaldehyde Resin involves the following steps: 1. Phenol, formaldehyde, water, and a catalyst are mixed in the desired amount, depending on the resin to be formed. 2. The mixture is then heated. The first part of the reaction, at around 70 °C, forms a thick reddish-brown tacky material, which is rich in hydroxymethyl and benzylic ether groups. 3. The concentration of the reactive form of formaldehyde depends on temperature and pH, as formaldehyde exists predominantly in solution as a dynamic equilibrium of methylene glycol oligomers.

Phenol formaldehyde resins were the first completely synthetic polymers to be commercialized and were used as the basis for Bakelite .

Preparation of Urea Formaldehyde (UF) Resin

Urea Formaldehyde (UF) resin is a type of thermosetting synthetic polymer that is commonly used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. **Reacting Urea and Formaldehyde:** The first step in the preparation of UF resin is the reaction of urea and formaldehyde in an aqueous solution. The molar ratio of urea to formaldehyde is typically between 1:1.5 and 1:2.5. The reaction is carried out at a pH of 7-8 and a temperature of 70-90°C.

2. **Condensation:** The next step is the condensation of the reaction mixture. This is done by heating the mixture to 90-100°C and maintaining the temperature for 30-60 minutes. During this time, the water in the mixture evaporates, and the urea-formaldehyde condensation reaction takes place.

3. **Cooling and Neutralization:** After the condensation reaction is complete, the mixture is cooled to room temperature. The pH of the mixture is then adjusted to 7-8 using an acid or a base.

4. **Addition of Fillers and Modifiers:** Fillers and modifiers are added to the UF resin to improve its properties. Common fillers include wood flour, cellulose, and clay, while common modifiers include melamine, ammonium chloride, and ammonium sulfate.

5. **Final Adjustments:** The final step in the preparation of UF resin is making any necessary adjustments to its properties. This can include adjusting the pH, viscosity, and solids content of the resin.

After these steps, the UF resin is ready for use in various applications. It can be used as an adhesive, a finish, or a molding compound, depending on the specific needs of the user.

Preparation of Adipic acid / Paracetamol

Adipic acid

Adipic acid is an organic compound with the formula $(CH_2)_4(COOH)_2$. It is the most important dicarboxylic acid from an industrial perspective, with about 2.5 billion kilograms of this white crystalline powder being produced annually, mainly as a precursor for the production of nylon .

Preparation

Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil. The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .

Traditional feedstock for the synthesis of adipic acid is benzene. Benzene is hydrogenated using Ni-Al_2O_3 catalyst at 370–800 psi to cyclohexane. This is oxidized using Co catalyst to a mixture of cyclohexanone and cyclohexanol .

Paracetamol

Paracetamol is a common pain reliever and fever reducer.

Preparation

Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulphuric acid as catalyst .

Paracetamol is available as 325-mg or 500-mg tablets, which may include calcium stearate or magnesium stearate, cellulose, docusate sodium and sodium benzoate or sodium lauryl sulfate, starch, hydroxypropyl methylcellulose, propylene glycol, sodium starch glycolate, polyethylene glycol and Red #40 .

Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the current passing through the solution.
4. The cell constant is determined by measuring the conductance of a solution of known conductivity.
5. The conductance of the unknown solution is then calculated by dividing the measured conductivity by the cell constant.
6. The cell conductance is expressed in units of Siemens (S).

7. Factors that affect the cell conductance of a solution include the concentration and mobility of the ions present, the temperature of the solution, and the nature of the solvent.

Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The reaction between an ester and water to form a carboxylic acid and an alcohol is known as hydrolysis.
3. The rate of hydrolysis of an ester can be determined by measuring the change in concentration of one of the reactants or products over time.
4. The rate constant can be calculated using the integrated rate law for a first-order reaction.
5. The rate constant can also be determined graphically by plotting the natural logarithm of the concentration of the reactant or product versus time.
6. The slope of the line is equal to the negative of the rate constant.
7. The rate constant can also be determined using the half-life of the reaction, which is the time required for the concentration of the reactant to decrease to half of its initial value.
8. The half-life of a first-order reaction is independent of the initial concentration of the reactant and is equal to the natural logarithm of 2 divided by the rate constant.
9. The rate constant can be determined experimentally by measuring the time required for the concentration of the reactant to decrease to half of its initial value and using the above equation to calculate the rate constant.
10. The rate constant of hydrolysis of esters is affected by factors such as temperature, pH, and the presence of catalysts.
11. The rate constant increases with increasing temperature according to the Arrhenius equation.
12. The pH of the solution can affect the rate of hydrolysis of esters by changing the concentration of the reactive species.
13. The presence of a catalyst can increase the rate of hydrolysis of esters by providing an alternative reaction pathway with a lower activation energy.
14. The rate constant of hydrolysis of esters can be used to predict the rate of reaction under different conditions and to design more efficient processes for the hydrolysis of esters.
15. The determination of the rate constant of hydrolysis of esters is an important tool in the study of reaction kinetics and the design of chemical processes.

Element Detection and Identification of Functional Groups in Organic Compounds

1. **Element Detection:** The detection of elements in organic compounds involves the determination of the presence of certain elements such as carbon, hydrogen, nitrogen, sulfur, and halogens.
2. **Carbon and Hydrogen:** The presence of carbon and hydrogen in an organic compound can be detected by heating the compound with copper oxide. The carbon in the compound is oxidized to carbon dioxide and the hydrogen is oxidized to water.
3. **Nitrogen:** The presence of nitrogen in an organic compound can be detected by the Lassaigne's test. In this test, the organic compound is fused with sodium metal. The nitrogen in the compound reacts with the sodium to form sodium cyanide, which can be detected by its reaction with iron(II) sulfate and sodium hydroxide.
4. **Sulfur:** The presence of sulfur in an organic compound can be detected by heating the compound with sodium. The sulfur in the compound reacts with the sodium to form sodium sulfide, which can be detected by its reaction with lead acetate.
5. **Halogens:** The presence of halogens in an organic compound can be detected by the Beilstein test. In this test, the organic compound is heated with copper wire. The halogens in the compound react with the copper to form copper halides, which can be detected by their characteristic color.
6. **Functional Groups:** Functional groups are specific groups of atoms within a molecule that are responsible for the characteristic chemical reactions of the molecule. The identification of functional groups in an organic compound is important for understanding its chemical behavior.
7. **Common Functional Groups:** Some common functional groups in organic compounds include hydroxyl (-OH), carbonyl (-CO-), carboxyl (-COOH), amino (-NH2), and nitro (-NO2) groups.
8. **Identification of Functional Groups:** The identification of functional groups in an organic compound can be done by a variety of chemical tests, such as the Tollens' test for aldehydes, the iodoform test for methyl ketones, and the Lucas test for alcohols.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and in-

terests of the students.
- It is important for students to carefully read the list of experiments and be prepared for any changes made by the instructor.
- Students should also communicate with the instructor if they have any questions or concerns about the experiments chosen.

Course Outcomes

Course outcomes are statements that describe the knowledge, skills, and abilities that students are expected to demonstrate upon completion of a course. These outcomes serve as a guide for both instructors and students in terms of what is expected to be achieved by the end of the course. Some key points to consider when developing course outcomes include:

1. **Specificity**: Course outcomes should be specific and clearly defined, outlining the knowledge and skills that students are expected to acquire.
2. **Measurability**: Outcomes should be measurable, meaning that there should be a way to assess whether or not students have achieved the desired outcome.
3. **Relevance**: Outcomes should be relevant to the course content and aligned with the overall goals of the course.
4. **Achievability**: Outcomes should be achievable within the scope of the course, taking into account the time and resources available.
5. **Student-centered**: Outcomes should be focused on what the student will be able to do, rather than what the instructor will teach.

In summary, course outcomes serve as a roadmap for both instructors and students, providing a clear picture of what is expected to be achieved by the end of the course. They should be specific, measurable, relevant, achievable, and student-centered.

Upon completion of the course the student should be able to:

- Demonstrate knowledge and understanding of the subject matter.
- Apply the concepts and principles learned in the course to solve problems.
- Analyze and evaluate information and arguments.
- Communicate effectively in writing and orally.
- Work effectively in a team.
- Demonstrate ethical and professional behavior.
- Engage in lifelong learning and professional development.

Course Outcomes Bloom's

Bloom's taxonomy is a model of cognitive skills used to classify educational learning objectives and is a helpful tool for the development of learning outcomes. It specifically targets knowledge, skills, or attitudes for change by seeking to increase knowledge (cognitive domain), develop skills (psychomotor domain), or develop emotional aptitude or balance (affective domain).

- Bloom's taxonomy primarily provides instructors with a focus for developing their course learning outcomes.
- It can be used to increase one's understanding of the educational process.
- Verbs at the appropriate course level are a key aspect of well-written learning outcomes. Bloom's Revised Taxonomy can be used to identify verbs that best align with the course level.
- The process outlined in the taxonomy provides a scaffolding around which instructors can design their course.
- Bloom's taxonomy is a powerful tool to help develop learning outcomes because it explains the process of learning: Before you can understand a concept, you must remember it. To apply a concept you must first understand it. In order to evaluate a process, you must have analyzed it.
- Bloom's Taxonomy is a hierarchical classification of the different levels of thinking, and should be applied when creating course objectives. Course objectives are brief statements that describe what students will be expected to learn by the end of the course.

Level

- Level is a term that can refer to a relative position or rank within a graded hierarchy.
- In the context of measurements, a level is a device used to determine if a surface is horizontal or vertical.
- In the context of video games, a level is a section or stage of the game that the player must complete to progress.
- In the context of education, a level can refer to a stage of academic progress, such as a grade level or degree level.
- In the context of sound, a level is a measure of the intensity or amplitude of a sound wave, often measured in decibels.
- In the context of liquids, a level is the height of the liquid's surface, often measured in relation to a fixed reference point.
- In the context of buildings, a level is a floor or storey within a multi-storey building.
- In the context of mathematics, a level is a value on a scale of measurement or a point on a graph.
- In the context of computer programming, a level can refer to the degree of abstraction or detail in a program or data structure.
- In the context of security, a level is a designation of the degree of protection

or access control applied to information or a system.

CO-1: Understanding the use of different analytical instruments

Analytical instruments are tools used to measure, analyze, and quantify physical and chemical properties of substances. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.
4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Thermal analysis instruments**: These instruments measure the thermal properties of a sample. They are commonly used to determine the melting point, boiling point, and heat capacity of a substance.

Understanding the use of these different analytical instruments is important for conducting accurate and reliable experiments. By selecting the appropriate instrument for a given experiment, researchers can obtain accurate and precise measurements of the properties of the substances they are studying.

CO2: Measuring Molecular/System Properties such as Surface Tension

Carbon dioxide (CO2) is a colorless, odorless gas that is naturally present in the Earth's atmosphere. It is an important greenhouse gas that helps regulate the temperature of the planet. However, human activities such as burning fossil fuels and deforestation have increased the concentration of CO2 in the atmosphere, contributing to global warming.

Surface tension is a property of liquids that arises from the cohesive forces between molecules at the surface of a liquid. It is the force that allows insects to walk on water and causes water droplets to form beads on a surface.

To measure the surface tension of CO2, several methods can be used, including:

1. The maximum bubble pressure method: This method involves measuring the pressure required to force a gas bubble through a liquid. The surface tension is calculated from the maximum pressure recorded.
2. The drop weight method: This method involves measuring the weight of drops of liquid that detach from the end of a thin tube. The surface tension is calculated from the weight of the drops and the size of the tube.
3. The Wilhelmy plate method: This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it. The surface tension is calculated from the force and the dimensions of the plate.

These are just a few of the methods that can be used to measure the surface tension of CO2. Each method has its advantages and limitations, and the choice of method will depend on the specific requirements of the experiment.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- High quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while sea water is about 50 mS/cm (or 0.05 S/cm).
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm.

Chloride and Iron Content in the Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.

- The presence of chloride ions in water can affect its conductivity, as mentioned above.
- Iron content in water can also affect its properties, but I couldn't find any specific information on this topic in my search results.

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3 surface: In mathematics, a K3 surface is a type of complex surface that has several interesting properties.
2. K3 (band): K3 is a Belgian-Dutch girl group that has been active since 1998.
3. K3 visa: In the United States, the K3 visa is a nonimmigrant visa for the spouse of a U.S. citizen.
4. K3 (mountain): K3, also known as Gasherbrum IV, is a mountain in the Karakoram range in Pakistan.

CO-3 Measure the hardness and alkalinity of the water

Hardness and alkalinity are two important parameters that are used to measure the quality of water. Here are some points to consider when measuring the hardness and alkalinity of water:

1. **Hardness** refers to the concentration of calcium and magnesium ions in water. These ions can cause problems in industrial and domestic water systems by forming scale deposits on pipes and equipment.

2. **Alkalinity** refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, which is important for maintaining a stable pH.

3. To measure the hardness of water, a titration method can be used. This involves adding a chemical reagent to a water sample until a color change occurs, indicating that all the calcium and magnesium ions have reacted.

4. To measure the alkalinity of water, a similar titration method can be used. In this case, a chemical reagent is added to the water sample until a pH change occurs, indicating that all the buffering capacity of the water has been used up.

5. It is important to measure both the hardness and alkalinity of water to ensure that it is suitable for its intended use. For example, water with

high hardness can cause problems in industrial processes, while water with low alkalinity can be corrosive to pipes and equipment.

6. The results of hardness and alkalinity tests can be used to determine the appropriate treatment methods for water, such as softening or pH adjustment.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that can be prepared through various methods. Some of the fundamental concepts of the preparation of phenol are:

1. **Preparation of Phenols from Haloarenes**: Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. **Preparation of Phenols from Diazonium Salts**: Phenols can also be prepared from diazonium salts.
3. **Preparation of Phenols from Benzene Sulphonic Acid**: At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
4. **Industrial Preparation of Phenols**: Phenol can be prepared industrially through various methods such as extraction from coal tar, Raschig's Process, and the air oxidation of cumene.
5. **Synthesis of Phenols**: Phenols can be synthesized in large quantities by the pyrolysis of the sodium salt of benzene sulfonic acid, by the Dow process, and by the air oxidation of cumene. Small amounts of phenol can also be prepared by the peroxide oxidation of phenylboronic acid and the hydrolysis of diazonium salts.

These are some of the fundamental concepts of the preparation of phenol. Phenol is an important chemical used in various industries and has many applications.

Formaldehyde & Urea Formaldehyde Resin

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer produced from urea and formaldehyde.
- UF is synthesized by the "alkali-acid-alkali" method .

- UF is used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.

Adipic Acid

- Adipic acid dihydrazide can be used as a modifier to reduce the free formaldehyde in UF resin.

Paracetamol

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3, a Belgian-Dutch girl group formed in 1998.
2. K3, a mountain in the Karakoram range, also known as Broad Peak.
3. K3, a type of visa for spouses of U.S. citizens to enter the United States while waiting for their immigrant visa to be processed.
4. K3, a type of high-speed train operated by China Railway High-speed.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k, is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law of the reaction.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the order of the reaction with respect to each reactant by performing experiments in which the concentration of one reactant is varied while the concentrations of the other reactants are kept constant.

2. Write the rate law for the reaction using the determined orders of reaction.

3. Measure the initial rate of the reaction for several sets of initial concentrations of the reactants.

4. Substitute the measured initial rates and the corresponding initial concentrations of the reactants into the rate law to obtain a set of simultaneous equations.

5. Solve the set of simultaneous equations to obtain the value of the rate constant, k.

It is important to note that the value of the rate constant, k, is dependent on the temperature at which the reaction is carried out. The relationship between the rate constant and the temperature is given by the Arrhenius equation. Therefore, to accurately estimate the rate constant of a reaction, the temperature at which the reaction is carried out must be carefully controlled and recorded.

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- Page 32 is the thirty-second page of a forty-page document.
- It is located after page 31 and before page 33.
- The content of page 32 depends on the specific document it is a part of.
- Without more information about the document, it is not possible to provide further details about the content of page 32.

ENGINEERING MATHEMATICS LAB

Engineering Mathematics Lab is a course designed to provide students with hands-on experience in applying mathematical concepts and techniques to solve engineering problems. The course typically covers topics such as:

1. Differential equations: Students learn to solve first and second-order differential equations using analytical and numerical methods.
2. Linear algebra: Students learn to solve systems of linear equations using matrix methods, and to perform operations such as matrix inversion and eigenvalue decomposition.
3. Probability and statistics: Students learn to apply statistical methods to analyze data, and to use probability theory to model uncertainty in engineering systems.
4. Numerical methods: Students learn to use numerical methods to approximate solutions to mathematical problems that cannot be solved analytically.

The course typically involves a combination of lectures, computer-based exercises, and group projects. Students are expected to use mathematical software such as MATLAB or Mathematica to complete assignments and projects.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance the ability to apply theoretical concepts to practical situations.
4. To encourage independent learning and research.
5. To foster effective communication and collaboration skills.

6. To prepare students for further academic or professional pursuits in the field.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in various fields such as chemistry, biology, medicine, and environmental science. Some of the fundamental concepts of analytical instruments include:

- **Accuracy and precision:** These terms refer to how close a measurement is to the true value (accuracy) and how consistent the measurements are (precision).
- **Sensitivity and selectivity:** Sensitivity refers to the ability of an instrument to detect small changes in the property being measured, while selectivity refers to the ability of an instrument to distinguish between different substances.
- **Calibration:** This is the process of adjusting an instrument to ensure that its measurements are accurate and precise.
- **Signal-to-noise ratio:** This refers to the ratio of the signal (the measurement of interest) to the noise (unwanted variations in the measurement).
- **Resolution:** This refers to the smallest difference between two measurements that can be distinguished by an instrument.

These are some of the fundamental concepts that students should understand when learning about analytical instruments. Understanding these concepts will enable students to use analytical instruments effectively and accurately in their work.

2. To enable the students to understand about the analysis of chloride content, hardness, alkalinity of water.

- **Chloride content:** Chloride is one of the major anions found in water and wastewater. The analysis of chloride content in water is important because high levels of chloride can be harmful to aquatic life and can also affect the taste of drinking water. The chloride content in water can be determined using various methods such as titration, ion chromatography, and ion-selective electrodes.

- **Hardness:** Hardness in water is caused by the presence of dissolved calcium and magnesium ions. It is important to analyze the hardness of water because it can affect the performance of water-using appliances and can also cause scaling in pipes and boilers. The hardness of water can be determined using various methods such as titration, atomic absorption spectroscopy, and ion exchange chromatography.

- **Alkalinity:** Alkalinity is a measure of the water's ability to neutralize acids. It is important to analyze the alkalinity of water because it can affect the pH of the water and can also affect the effectiveness of water treatment processes. The alkalinity of water can be determined using various methods such as titration, colorimetry, and ion chromatography.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 is acidic, while a pH greater than 7 is basic.
- **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the molecules of the liquid. Surface tension can be measured using a variety of methods, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
- **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity can be measured using a variety of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method.

4. To enable the students to understand about the preparation of different resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to help students understand the preparation of different resins:

1. **Natural resins**: Natural resins are typically obtained from plants. For example, pine resin can be collected by tapping the trunk of pine trees. The resin is then purified and used for various purposes.

2. **Synthetic resins**: Synthetic resins are prepared by the polymerization of monomers. For example, phenol-formaldehyde resin is prepared by the reaction of phenol and formaldehyde in the presence of an acid or base catalyst.

3. **Curing**: Many resins require curing, which is the process of hardening the resin by the application of heat, pressure, or a combination of both. Curing can also be achieved through the use of a catalyst or by the addition of a curing agent.

4. **Additives**: Additives can be added to resins to modify their properties. For example, plasticizers can be added to increase the flexibility of the resin, while fillers can be added to improve the strength and stiffness of the resin.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol by Conventional and Green Route

Adipic acid and paracetamol are two important organic compounds that can be synthesized through both conventional and green routes. Here, we will discuss the synthesis of these compounds through both methods.

Adipic Acid

Adipic acid is an important dicarboxylic acid used in the production of nylon. It can be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of adipic acid involves the oxidation of cyclohexanol or cyclohexanone with nitric acid. This process produces nitrous oxide, a potent greenhouse gas, as a byproduct.

Green Route

The green route for the synthesis of adipic acid involves the oxidation of cyclohexene with hydrogen peroxide in the presence of a tungsten catalyst. This process produces water as the only byproduct, making it a more environmentally friendly option.

Paracetamol

Paracetamol is a widely used analgesic and antipyretic drug. It can also be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of paracetamol involves the nitration of phenol to produce para-nitrophenol, which is then reduced to para-aminophenol. This is then acetylated to produce paracetamol.

Green Route

The green route for the synthesis of paracetamol involves the direct acetylation of para-aminophenol with acetic anhydride in the presence of a solid acid catalyst. This process produces fewer byproducts and is considered to be more environmentally friendly.

In conclusion, the synthesis of organic compounds such as adipic acid and paracetamol can be achieved through both conventional and green routes. The green routes are considered to be more environmentally friendly due to the production of fewer byproducts and the use of less harmful reagents. It is important for students to understand the differences between these routes and the benefits of using green chemistry in the synthesis of organic compounds.

LIST OF EXPERIMENTS

- List experiments are a questionnaire design technique used to mitigate respondent social desirability bias when eliciting information about sensitive topics. With a large enough sample size, list experiments can be used to estimate the proportion of people for whom a sensitive statement is true.
- In a list experiment, the sample is randomly divided into two groups: the Direct Response Group and the Veiled Response Group.
- Some famous experiments include Harry Harlow's experiments with baby monkeys and wire and cloth surrogate mothers (1957–1974), Stanley Milgram's experiments on human obedience (1963), and Walter Mischel's marshmallow experiment showing the importance to life outcomes of the ability to delay gratification (beginning late 1960s).
- There are also many science experiments that can be done at home, such as kitchen chemistry and DIY mini drones .

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing it with a reference standard. In the context of analytical equipment and apparatus, calibration is essential to ensure that the measurements obtained are accurate and reliable.

Here are some key points to consider when calibrating analytical equipment and apparatus:

- Calibration should be performed regularly to ensure the accuracy of the instrument over time.
- The reference standard used for calibration should be traceable to an internationally recognized standard.
- The calibration process should be documented, including the date of calibration, the reference standard used, and the results obtained.
- Any deviations from the expected results should be investigated and corrective actions taken if necessary.
- The calibration process should be performed by trained personnel to ensure that it is carried out correctly.

In summary, calibration of analytical equipment and apparatus is an essential step in ensuring the accuracy and reliability of measurements. It should be performed regularly, using traceable reference standards, and the process should be documented and carried out by trained personnel.

2. Determination of Hardness of water sample by EDTA method

The hardness of water is due to the presence of calcium and magnesium salts in water. The EDTA method is one of the three main methods for determining the hardness of water. EDTA stands for Ethylenediaminetetraacetic acid. This reagent can form EDTA-metal complexes by reacting with metal ions, except for alkali metal ions .

The procedure for calculating the hardness of water by EDTA titration is as follows :

1. Take a sample volume of 20ml (V ml).
2. Dilute 20ml of the sample in an Erlenmeyer flask to 40ml by adding 20ml of distilled water.
3. Add 1 mL of ammonia buffer to bring the pH to 10±0.1.
4. Add 1 or 2 drops of the indicator solution.
5. Add EDTA titrant to the sample with vigorous shaking until the wine red color just turns blue.

The hardness can then be calculated using the formula: Hardness (EDTA), as mg/L = × ×1,000 where A = mL of EDTA titrant used, T = Titer of EDTA titrant, mg CaCO3 per mL of EDTA titrant, and S = mL of sample volume . The result can be reported as "Total Hardness (EDTA) = ______ mg/L as CaCO3" or as "Calcium Hardness (EDTA) = ______ mg/L as CaCO3" .

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the ability of water to neutralize acids. It is determined by measuring the concentration of bicarbonate, carbonate, and hydroxide ions in a water sample. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter.
3. Add a few drops of phenolphthalein indicator to the water sample. If the pH is above 8.3, the water will turn pink, indicating the presence of carbonate ions.
4. Titrate the water sample with a standard solution of sulfuric acid or hydrochloric acid until the pink color disappears. Record the volume of acid used.
5. Calculate the alkalinity of the water sample using the following formula: Alkalinity (mg/L as CaCO3) = (Volume of acid used x Normality of acid x 50,000) / Volume of water sample.

It is important to note that the alkalinity of water can vary depending on the source and the treatment processes it has undergone. Alkalinity is an important parameter in water treatment and can affect the effectiveness of disinfection and the stability of the water distribution system.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the indicator changes color. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where precise pH measurements are required.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height of the liquid that rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: `Surface tension = ( ghr)/2`, where   is the density of the liquid, `g` is the acceleration due to gravity, `h` is the height of the liquid column, and `r` is the radius of the capillary tube.

2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble from the end of a capillary tube immersed in the liquid. The surface tension is then calculated using the formula: `Surface tension = (4Pm)/d`, where `Pm` is the maximum pressure and `d` is the diameter of the capillary tube.

3. **Drop weight method:** This method involves measuring the weight of a drop of liquid that detaches from the end of a capillary tube. The surface tension is then calculated using the formula: `Surface tension = mg/ d`, where `m` is the mass of the drop, `g` is the acceleration due to gravity, and `d` is the diameter of the capillary tube.

4. **Wilhelmy plate method:** This method involves immersing a thin flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: `Surface tension = F/L`, where `F` is the force and `L` is the perimeter of the plate.

5. **Du Nouy ring method:** This method involves immersing a ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: `Surface tension = (F - Fc)/2 r`, where `F` is the force, `Fc` is the correction factor, and `r` is the radius of the ring.

These are some of the common methods used to determine the surface tension of a given liquid. The choice of method depends on the accuracy required and the availability of equipment.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. A viscometer is an instrument used to measure the viscosity of a fluid.

To determine the viscosity of a given liquid using a viscometer, the following steps can be followed:

1. Fill the viscometer with the liquid to be tested.
2. Place the viscometer in a constant temperature bath to ensure that the temperature of the liquid remains constant during the experiment.
3. Allow the liquid to flow through the viscometer and measure the time it takes for a known volume of liquid to flow through the viscometer.
4. Repeat the experiment several times to obtain an average value for the flow time.
5. Use the flow time and the known dimensions of the viscometer to calculate the viscosity of the liquid using the appropriate formula.

It is important to note that the viscosity of a liquid is dependent on its temperature, so it is important to control the temperature during the experiment. Additionally, the viscometer should be calibrated before use to ensure accurate results.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate. It is commonly used in the determination of the strength of oxidizing agents using an external indicator.

1. The strength of ferrous ammonium sulfate can be determined by titrating it against a standard solution of potassium permanganate (KMnO4) using an external indicator such as potassium dichromate (K2Cr2O7).
2. The reaction between ferrous ammonium sulfate and potassium permanganate is an oxidation-reduction reaction, where the iron (II) ions in ferrous ammonium sulfate are oxidized to iron (III) ions by the permanganate ions.

3. The end point of the titration is indicated by the change in color of the external indicator, potassium dichromate, from orange to green.
4. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the molar concentration of the potassium permanganate solution.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is diphenylamine sulfonic acid.
4. The end point is indicated by the appearance of a blue color.
5. The strength of the potassium dichromate solution can be calculated using the formula: Normality of potassium dichromate = (Normality of ferrous ammonium sulfate x Volume of ferrous ammonium sulfate) / Volume of potassium dichromate.
6. The experiment should be repeated to obtain an average value for the strength of the potassium dichromate solution.

9. Determination of available chlorine in bleaching powder

1. The available chlorine present in a bleaching powder sample is determined iodometrically by treating its solution with an excess of potassium iodide solution in an acidic medium.
2. The liberated iodine (I2) is treated with sodium thiosulphate (Na2S2O3) solution using freshly prepared starch solution as an indicator to be added near the endpoint.
3. Bleaching powder is commonly used as a disinfectant. The chlorine present in the bleaching powder gets reduced with time.
4. The chlorine that is liberated is called available chlorine, and the calculation of available chlorine is called iodometry.
5. Available chlorine is determined by adding a measured amount of sample solution to an acidified solution of potassium iodide, liberating an equivalent amount of iodine. This free iodine is determined by titration with standardized sodium thiosulfate solution, using a starch indicator endpoint.

10. Determination of chloride content in water sample

1. Chloride is one of the major anions found in water and wastewater.
2. The chloride content of water can be determined by titration with silver nitrate solution using potassium chromate as an indicator.
3. The reaction between silver nitrate and chloride ions produces a white precipitate of silver chloride.

4. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate.
5. The concentration of chloride ions in the water sample can be calculated from the volume of silver nitrate solution used in the titration.
6. The presence of high levels of chloride in water can indicate pollution from sources such as sewage, industrial effluents, and road salt.
7. The determination of chloride content in water is important for monitoring the quality of drinking water, as well as for the control of industrial processes and wastewater treatment.
8. The method used for the determination of chloride content in water is standardized and described in various analytical methods, such as the Standard Methods for the Examination of Water and Wastewater.
9. The accuracy of the determination of chloride content in water can be affected by various factors, such as the presence of interfering substances and the precision of the titration.
10. Quality control measures, such as the use of standard solutions and the analysis of duplicate samples, can help to ensure the accuracy and reliability of the determination of chloride content in water.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed**: The first step in the preparation of PF resin is to mix phenol and formaldehyde in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol is typically between 1.5:1 and 2:1.

2. **Formation of hydroxymethylphenols**: The reaction between phenol and formaldehyde results in the formation of hydroxymethylphenols, which can further react with each other or with more formaldehyde to form more complex compounds.

3. **Formation of resin**: As the reaction progresses, the hydroxymethylphenols react with each other to form larger molecules, eventually resulting in the formation of a resin. The resin can be either in the form of a liquid or a solid, depending on the specific conditions of the reaction.

4. **Curing**: The final step in the preparation of PF resin is curing, which involves heating the resin to a high temperature to complete the polymerization process. This results in a hard, durable material that is resistant to heat and chemicals.

PF resin is widely used in the production of various products, including laminates, adhesives, and molded objects. It is known for its high strength, durability, and resistance to heat and chemicals.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin is a type of thermosetting polymer that is widely used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. The first step in the preparation of UF resin is the reaction between urea and formaldehyde. This reaction takes place in an aqueous solution, with the molar ratio of urea to formaldehyde typically ranging from 1:1.5 to 1:2.5.
2. The reaction is carried out under slightly acidic conditions, with a pH of around 5, and at a temperature of 60-70°C.
3. The reaction between urea and formaldehyde produces a number of intermediate compounds, including monomethylol urea and dimethylol urea.
4. These intermediate compounds then undergo further condensation reactions to form a viscous, water-soluble resin.
5. The resin is then concentrated by removing excess water through evaporation or filtration.
6. The concentrated resin can then be used as-is or further modified by the addition of fillers, pigments, or other additives to achieve the desired properties.
7. The final step in the preparation of UF resin is the curing process, which involves heating the resin to a temperature of 120-150°C in order to crosslink the polymer chains and form a rigid, thermoset material.

It is important to note that the exact conditions and proportions used in the preparation of UF resin can vary depending on the desired properties of the final product. Additionally, the use of catalysts or other additives can also influence the reaction and the properties of the resulting resin.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and Paracetamol are two different compounds with different preparation methods.

Adipic acid:

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is a white crystalline powder and is mainly used in the production of nylon.

1. Adipic acid can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
2. Another method involves the oxidation of cyclohexane with air in the presence of a cobalt catalyst.

3. Adipic acid can also be prepared from benzene through a series of reactions involving the formation of cyclohexene and its subsequent oxidation.

Paracetamol:

Paracetamol, also known as acetaminophen, is an over-the-counter pain reliever and fever reducer. It is commonly used to treat headaches, menstrual cramps, toothaches, and other minor aches and pains.

1. Paracetamol can be prepared by the nitration of phenol to produce para-nitrophenol, which is then reduced to para-aminophenol.
2. The para-aminophenol is then acetylated with acetic anhydride to produce paracetamol.
3. Another method involves the direct acetylation of phenol with acetic anhydride in the presence of a catalyst such as zinc chloride.

14. Determination of Cell Conductance of a solution

1. Cell conductance is a measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the current flowing between the electrodes.
4. The cell is filled with the solution to be tested and the conductivity is measured.
5. The cell constant is determined by measuring the conductivity of a standard solution of known conductivity.
6. The cell conductance is calculated by dividing the measured conductivity by the cell constant.
7. The cell conductance is affected by the concentration and mobility of the ions in the solution, as well as the temperature of the solution.
8. The cell conductance can be used to determine the concentration of ions in a solution, or to monitor changes in the concentration of ions over time.

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction between an ester and water.
2. The reaction is typically carried out in the presence of an acid or base catalyst to increase the rate of reaction.
3. The rate of reaction can be measured by monitoring the change in concentration of one of the reactants or products over time.
4. The rate constant can then be calculated using the integrated rate law for a first-order reaction.
5. The rate constant is dependent on temperature and can be determined at different temperatures to calculate the activation energy of the reaction.

6. The rate constant can also be affected by the nature of the ester and the catalyst used.
7. The hydrolysis of esters is an important reaction in organic chemistry and has many practical applications, such as in the production of soaps and detergents.

16. Element detection and identification of functional groups in organic compounds

1. **Element detection** in organic compounds involves the determination of the presence of certain elements such as carbon, hydrogen, nitrogen, sulfur, and halogens.
2. **Functional groups** are specific groups of atoms within molecules that are responsible for the characteristic chemical reactions of those molecules.
3. The identification of functional groups in organic compounds can be achieved through various chemical tests and spectroscopic techniques.
4. Some common functional groups include alcohols, aldehydes, ketones, carboxylic acids, esters, amines, and amides.
5. The presence of certain functional groups can be determined through characteristic chemical reactions. For example, the presence of a carboxylic acid functional group can be determined through its reaction with sodium bicarbonate to produce carbon dioxide gas.
6. Spectroscopic techniques such as infrared spectroscopy and nuclear magnetic resonance spectroscopy can also be used to identify functional groups in organic compounds.
7. Infrared spectroscopy involves the measurement of the absorption of infrared radiation by a molecule, which can provide information about the types of bonds and functional groups present in the molecule.
8. Nuclear magnetic resonance spectroscopy involves the measurement of the absorption of radiofrequency radiation by the nuclei of atoms in a molecule, which can provide information about the chemical environment of the nuclei and the functional groups present in the molecule.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided above.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions for each experiment.
- Students should also be prepared for the possibility that the instructor may change some of the experiments.

- It is recommended that students communicate with the instructor to clarify any questions or concerns they may have about the experiments.

Course Outcomes:

- A course outcome is a statement that describes the knowledge, skills, and abilities that students should possess upon completion of a course.
- Course outcomes are used to guide the development of course content, assessments, and teaching methods.
- Course outcomes should be specific, measurable, and achievable within the scope of the course.
- Course outcomes should align with program outcomes and institutional goals.
- Course outcomes should be regularly reviewed and updated to ensure their relevance and effectiveness.
- Course outcomes can be used to evaluate the effectiveness of a course and to identify areas for improvement.
- Course outcomes can be used to communicate expectations to students and to provide a framework for assessing student learning.
- Course outcomes can be used to facilitate the transfer of credit and to demonstrate the value of a course to external stakeholders.

Upon completion of the course the student should be able to:

1. Demonstrate a thorough understanding of the course material.
2. Apply the concepts and theories learned in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.
4. Communicate ideas and arguments clearly and effectively in both written and oral forms.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior in all course-related activities.
7. Use technology effectively to enhance learning and communication.
8. Engage in continuous learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is commonly used to design course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering**: The ability to recall or recognize information.
2. **Understanding**: The ability to comprehend the meaning of information.
3. **Applying**: The ability to use information in a new situation.
4. **Analyzing**: The ability to break down information into its component parts and understand their relationships.
5. **Evaluating**: The ability to make judgments about the value of information.
6. **Creating**: The ability to combine information to form a new whole.

Course outcomes can be written using verbs from Bloom's Taxonomy to indicate the level of thinking required by the student. For example, a course outcome for a history course might be: "Students will be able to analyze the causes and consequences of major historical events." This outcome uses the verb "analyze" from the Analyzing level of Bloom's Taxonomy.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels are used in construction, carpentry, surveying, and many other applications where accurate measurements are required.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness. The bubble will be centered between two lines when the surface is level.
- Laser levels project a laser beam to create a straight line on a surface. They can be used to determine levelness over a longer distance than a spirit level.
- Water levels use the principle that water will always find its own level. They consist of a length of tubing filled with water, with a clear section at each end. When the two ends are held at the same height, the water level in the clear sections will be the same, indicating that the two points are level with each other.
- Levels are an essential tool for ensuring accuracy in many different types of projects. They are widely available and easy to use.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are used to measure physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or

transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.

3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.

4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Microscopes**: These instruments magnify small objects to make them visible to the naked eye. They are commonly used to study the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations. Understanding how to use these instruments and interpret their results is essential for conducting accurate and reliable analyses.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a physical property of liquids that results from the cohesive forces between the molecules at the surface of a liquid. It is defined as the force acting per unit length on an imaginary line drawn on the liquid surface, tending to pull the surface molecules together.

Some key points to remember about surface tension are:

1. Surface tension is caused by the imbalance of intermolecular forces at the surface of a liquid.
2. The units of surface tension are typically expressed in N/m or dynes/cm.
3. Surface tension can be measured using various methods such as the maximum bubble pressure method, the drop weight method, and the Wilhelmy plate method.
4. Surface tension is temperature-dependent and typically decreases with increasing temperature.
5. Surfactants can be used to reduce the surface tension of a liquid.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

- **Viscosity** is a measure of a fluid's resistance to flow. It describes the internal friction of a moving fluid. A fluid with high viscosity resists motion because its molecular makeup gives it a lot of internal friction. A fluid with low viscosity flows easily because its molecular makeup results in very little friction when it is in motion.

- **Conductance of a solution** is the ability of a solution to conduct an electric current. The more ions present in the solution, the higher the conductivity. For example, the conductivity of seawater, which contains a high concentration of ions, is much higher than that of pure water, which contains very few ions. The conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm .

- **Chloride content in water** refers to the amount of chloride ions present in the water. Chloride ions are approximately 1.9 percent of the mass of seawater . The more chloride ions present in the water, the higher the conductivity .

- **Iron content in water** refers to the amount of iron present in the water. Iron can be present in water in two forms: soluble ferrous iron or insoluble ferric iron. Water containing ferrous iron is clear and colorless because the iron is completely dissolved. When exposed to air in the pressure tank or atmosphere, the water turns cloudy and a reddish brown substance begins to form. This sediment is the oxidized or ferric form of iron that will not dissolve in water.

K3

- K3 is a term that can refer to several different things, depending on the context.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a type of visa for spouses of U.S. citizens.
- In mathematics, K3 can refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters of water quality. They are used to determine the suitability of water for various uses, such as drinking, irrigation, and industrial processes.

- **Hardness** refers to the concentration of calcium and magnesium ions in water. These ions can form insoluble compounds with soap, causing it to lose its cleaning power. Hard water can also cause scaling in pipes and appliances, reducing their efficiency.

- **Alkalinity** refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, which is determined by the presence of bicarbonate, carbonate, and hydroxide ions. Alkaline water can help to neutralize acidic waste products in the body, and is believed to have health benefits.

To measure the hardness of water, a sample is collected and titrated with a standard solution of ethylenediaminetetraacetic acid (EDTA). The amount of EDTA required to react with all the calcium and magnesium ions in the sample is used to calculate the hardness of the water.

To measure the alkalinity of water, a sample is collected and titrated with a standard solution of sulfuric acid. The amount of acid required to lower the pH of the sample to a specific value is used to calculate the alkalinity of the water.

Both hardness and alkalinity can be expressed in various units, such as milligrams per liter (mg/L) or grains per gallon (gpg). The appropriate units to use depend on the intended use of the water and the standards set by regulatory agencies.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a white crystalline solid that is volatile and has a sweet, tarry odor. Phenol is used in the production of many different chemicals, including plastics, resins, and pharmaceuticals.

There are several methods for the preparation of phenol, including:

1. **Cumene process:** This is the most common method for the production of phenol. In this process, cumene is oxidized to cumene hydroperoxide, which is then cleaved to produce phenol and acetone.

2. **Raschig-Hooker process:** In this process, chlorobenzene is reacted with sodium hydroxide to produce phenol and sodium chloride.

3. **Dow process:** This process involves the reaction of chlorobenzene with caustic soda at high temperatures and pressures to produce phenol and sodium chloride.

4. **From benzene:** Phenol can also be produced by the direct hydroxylation of benzene using hydrogen peroxide and a catalyst.

These are some of the fundamental concepts of the preparation of phenol. It is important to understand these methods in order to have a better understanding of the production of this important chemical.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.

- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer produced from urea and formaldehyde.
- UF is synthesized by the "alkali-acid-alkali" method using formaldehyde and urea as raw materials .
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .
- UF resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Melamine, polyvinyl alcohol, and adipic acid dihydrazide can be added to UF resin as modifiers for reducing the free formaldehyde.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a type of visa for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- In the context of technology, K3 could refer to a model of a product, such as a keyboard or a camera.
- Without more information, it is difficult to determine the specific meaning of K3 in this context.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k, is a measure of the speed at which a chemical reaction occurs. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective orders. The units of the rate constant depend on the order of the reaction.

There are several methods to estimate the rate constant of a reaction, including:

1. **Graphical methods**: A graph of the concentration of a reactant or product versus time can be used to determine the order of the reaction and the rate constant. For example, for a first-order reaction, a plot of the natural logarithm of the concentration versus time will give a straight line with a slope equal to -k.

2. **Initial rates method**: This method involves measuring the initial rate of the reaction at different initial concentrations of the reactants. The data can then be used to determine the order of the reaction and the rate constant.

3. **Half-life method**: For a first-order reaction, the half-life is defined as the time required for the concentration of a reactant to decrease to half of its

initial value. The half-life is related to the rate constant by the equation t1/2 = ln(2)/k.

4. **Integrated rate laws**: The integrated rate laws for zeroth-, first-, and second-order reactions can be used to determine the rate constant by fitting experimental data to the appropriate equation.

It is important to note that the rate constant is temperature-dependent and can be affected by factors such as the presence of a catalyst. The Arrhenius equation can be used to determine the temperature dependence of the rate constant.

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- Is this page part of a book, document, or another type of resource?
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ENGINEERING GRAPHICS LAB

Engineering Graphics Lab is an important subject of engineering which provides students an opportunity to enhance their capability of imagination and lateral thinking . It offers students an insight into the methods to tackle engineering problems. It teaches us the principle of accuracy, exactness & positiveness with regards to the information necessary for the production of the engineering components .

The Engineering Design Graphics Labs consist of student workstations plus one teacher station. These labs are used specifically for Engineering Graphics, Computer Aided Engineering, and Manufacturing courses. Each computer has multiple software packages installed, including SolidWorks, AutoCAD, Microstation .

Engineering drawing is a means of communication. Principals of engineering graphics include free hand sketching, and introduction to AutoCAD and AutoCAD commands. Engineering drawing with AutoCAD; orthographic projection; lines and dimensioning; reading blueprints; normal, inclined and cylindrical surfaces; sectional views .

Drafting methods & techniques include multiview orthographic projection, isometrics and pictorials, sections & auxiliary views, dimensioning and tolerancing. Drafting and sketching using instruments and CAD .

Course Objectives

- To provide a comprehensive understanding of the subject matter.
- To develop critical thinking and analytical skills.
- To enhance problem-solving abilities.
- To encourage independent learning and research.
- To foster effective communication and collaboration.
- To promote the application of theoretical knowledge to real-world scenarios.
- To prepare students for further academic or professional pursuits in the field.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices that are used to measure the physical and chemical properties of substances. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some of the fundamental concepts of analytical instruments include:

1. **Accuracy**: This refers to how close the measurement is to the true value. Analytical instruments must be accurate in order to provide reliable results.

2. **Precision**: This refers to how consistent the measurements are when the same sample is measured multiple times. Precision is important in order to ensure that the results are repeatable.

3. **Sensitivity**: This refers to the ability of the instrument to detect small changes in the sample. High sensitivity is important in order to detect small concentrations of substances.

4. **Selectivity**: This refers to the ability of the instrument to distinguish between different substances in a sample. Selectivity is important in order to accurately identify the substances present in a sample.

5. **Calibration**: This refers to the process of adjusting the instrument to ensure that it is providing accurate and precise measurements. Calibration is important in order to maintain the accuracy and precision of the instrument over time.

These are some of the fundamental concepts of analytical instruments that students should understand in order to effectively use these devices in their studies and research.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content**: Chloride content refers to the amount of chloride ions present in a water sample. Chloride ions are negatively charged particles that can be found in many types of water, including seawater, groundwater, and surface water. The analysis of chloride content is important because high levels of chloride can indicate contamination from sources such as sewage, industrial waste, or road salt.
2. **Hardness**: Water hardness refers to the amount of dissolved minerals, primarily calcium and magnesium, present in a water sample. Hard water can cause problems such as scaling in pipes and appliances, and can interfere with the effectiveness of soaps and detergents. The analysis of water hardness is important for determining the need for water softening or treatment.
3. **Alkalinity**: Alkalinity refers to the ability of water to neutralize acids. It is primarily a measure of the amount of bicarbonate, carbonate, and hydroxide ions present in a water sample. Alkalinity is important because it helps to regulate the pH of water, and can affect the solubility of certain minerals and the effectiveness of disinfectants. The analysis of alkalinity is important for maintaining the proper chemical balance of water.

These are some of the key concepts that students should understand when studying the analysis of chloride content, hardness, and alkalinity in water. It is important for students to have a strong grasp of these concepts in order to effectively analyze and interpret water quality data.

Water

- Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states.
- It is one of the most plentiful and essential of compounds.
- A tasteless and odourless liquid at room temperature, it has the important ability to dissolve many other substances.
- Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen.
- Water molecules cling to each other because of a force called hydrogen bonding.
- It exists in three states on Earth: liquid, gas, and solid.
- Water is the liquid that makes life on Earth possible.
- All living things, from tiny cyanobacteria to giant blue whales, need water to survive.
- Water is an inorganic compound with the chemical formula H2O.

- It is a transparent, tasteless, odorless, and nearly colorless chemical substance.
- It is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).

Measure of pH, Surface Tension, and Viscosity

1. **pH** is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 indicates an acidic solution, while a pH greater than 7 indicates a basic solution.

2. **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the molecules of the liquid. Surface tension can be measured using various methods, such as the maximum bubble pressure method or the drop weight method.

3. **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity can be measured using various methods, such as the capillary tube method or the rotating spindle method.

These three properties are important in various fields, such as chemistry, physics, and engineering. Understanding how to measure and interpret these properties can provide valuable insights into the behavior of fluids and solutions.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to conform to the shape of its container while retaining a constant volume.
- Liquids have a definite volume, but no definite shape.
- The molecules in a liquid are in constant motion and are loosely packed, allowing them to slide past one another.
- The attractive forces between the molecules in a liquid are strong enough to keep the molecules close together, but not strong enough to keep them in a fixed position.
- Liquids have a higher density than gases, but a lower density than solids.
- The temperature at which a substance changes from a solid to a liquid is called its melting point, and the temperature at which it changes from a liquid to a gas is called its boiling point.
- Liquids can be classified as either Newtonian or non-Newtonian, depending on their flow behavior.

- A Newtonian liquid has a constant viscosity, meaning its resistance to flow remains the same regardless of the applied stress.
- A non-Newtonian liquid, on the other hand, has a variable viscosity, meaning its resistance to flow changes depending on the applied stress.
- Examples of liquids include water, milk, blood, and oil.

Preparation of Different Resins

Resins can be derived from various sources such as polymers, monomers, oils, fatty acids, and natural sources. They have a wide range of applications in industries such as paints and coatings, personal care and home care, pharmaceuticals, rubber, plastics, electrical and jewelry articles, decorative items, wood coatings, etc.

There are two categories of resinous products: natural resins and prepared resins. So far, no general method has been suggested or proposed for the preparation of resins.

In resin casting, fluid synthetic resin is blended with a curing agent, commonly at room temperature or close to room temperature. The two substances are then poured into a mold cavity. The curing agent then converts the resin into solid polymers, essentially solidifying it.

The structure of the linker can be manipulated to prepare resins ranging from extremely acid labile to base labile. Using linkers additionally allows the preparation of resins with special applications such as DHP resin utilized as a solid-phase support for alcohols or Weinreb resin utilized for preparing aldehydes and ketones.

There are several types of resins, including epoxy resin, UV resin, polyester resin, and polyurethane resin. The majority of resins are made up of two elements: the base resin and a catalyst (which is a hardener). Combining the two elements results in a chemical reaction that allows the resin to set.

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other industrial products. Here are some key points to understand about the synthesis of adipic acid:

1. Adipic acid is a dicarboxylic acid with the chemical formula C6H10O4.
2. It can be synthesized from a variety of starting materials, including cyclohexanol, cyclohexanone, and benzene.
3. One common method for synthesizing adipic acid is through the oxidation of cyclohexanol or cyclohexanone using nitric acid.
4. Another method involves the oxidation of benzene using a mixture of nitric and sulfuric acids.

5. The synthesis of adipic acid can also be achieved through the use of biocatalysts, such as enzymes or microorganisms.
6. The choice of synthesis method depends on factors such as cost, availability of starting materials, and environmental considerations.

Acid and Paracetamol

- Mefenamic Acid + Paracetamol is a combination of two medicines: Mefenamic Acid and Paracetamol. It works by blocking the release of certain chemical messengers that cause fever, pain and inflammation (redness and swelling).
- Mefenamic Acid+Paracetamol is used for pain relief and fever. It relieves pain and inflammation in conditions like rheumatoid arthritis, osteoarthritis and menstrual pain. It is also used to treat mild to moderate pain.
- MEFENAMIC ACID+PARACETAMOL belongs to a group of medicines called Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) used for the treatment of pain, inflammation, migraine headache, period pain, heavy bleeding during periods, muscle pain, tooth pain, joint pain, pain after surgery, ear pain, fever, flu, osteoarthritis, and rheumatoid arthritis.
- Paracetamol (Panadol, Calpol, Alvedon) is an analgesic and antipyretic drug that is used to temporarily relieve mild-to-moderate pain and fever. It is commonly included as an ingredient in cold and flu medications and is also used on its own. Paracetamol is exactly the same drug as acetaminophen (Tylenol).
- Paracetamol is used as an analgesic and antipyretic drug. It is the preferred alternative analgesic-antipyretic to aspirin (acetylsalicylic acid), particularly in patients with coagulation disorders, individuals with a history of peptic ulcer or who cannot tolerate aspirin, as well as in children.